EMOTION RECOGNITION USING MULTIMODAL DEEP LEARNING WITH VISION TRANSFORMERS AND GRAPH-BASED POSE ANALYSIS

A Project Report

Submitted to the Faculty of Engineering of

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA, KAKINADA

In partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY
In
CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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CERTIFICATE

This is to certify that the project report entitled **"EMOTION RECOGNITION USING MULTIMODAL DEEP LEARNING WITH VISION TRANSFORMERS AND GRAPH-BASED POSE ANALYSIS"** is a Bonafide record of work carried out by **Betha Hari Srimukhi (22481A4215), Addada Nithya Phani Sri (22481A4201), Gutta Alekhya (22481A4235),** under the guidance and supervision of **Mrs. G. Kalyani** , **Assistant Professor , Department of CSE (Artificial Intelligence & Machine Learning)** in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in CSE(Artificial Intelligence and Machine Learning) of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2025- 26.

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of people who made it possible and whose constant guidance and encouragements crown all the efforts with success.

We would like to express our deep sense of gratitude and sincere thanks to **Mrs. G. Kalyani,** Assistant Professor, Department of CSE (Artificial Intelligence and Machine Learning) for his/her constant guidance, supervision and motivation in completing the project work.

We feel elated to express our floral gratitude and sincere thanks to **Dr. Y. Adilakshmi**, Head of the Department, CSE (Artificial Intelligence and Machine Learning) for her encouragements all the way during analysis of the project. Her annotations, insinuations and criticisms are the key behind the successful completion of the project work.

We would like to take this opportunity to thank our beloved principal **Dr. B. Karuna Kumar** for providing a great support for us in completing our project and giving us the opportunity for doing project.

Our Special thanks to the faculty of our department, non-teaching staff and . family members who had directly or indirectly helped and supported us in completing our project in time.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
ViT	Vision Transformer
PGCN	Pose Graph Convolutional Network
GCN	Graph Convolutional Network
HCA	Hierarchical Cross Attention
RAF-DB	Real-world Affective Faces Database
GPU	Graphics Processing Unit
API	Application Programming Interface
UI	User Interface
DL	Deep Learning
TP	True Positive
FP	False Positive
FN	False Negative
F1-score	Harmonic mean of precision and recall

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ABSTRACT

Emotion recognition plays an important role in understanding human behavior and improving interaction between humans and intelligent systems. It has applications in areas such as human–computer interaction, healthcare monitoring, smart classrooms, and intelligent surveillance. Traditional emotion recognition systems mainly rely on facial expressions and convolutional neural networks (CNNs). However, these approaches often struggle to perform reliably when facial information is unclear due to occlusion, lighting variations, or pose changes.

To address these limitations, this project proposes a **multimodal emotion recognition system** that combines facial expression analysis with body posture analysis. Facial features are extracted using a **Vision Transformer (ViT)** model, which captures global contextual relationships across facial regions through a self-attention mechanism. In addition to facial analysis, body posture information is obtained by extracting human pose keypoints using **MediaPipe Pose**. These keypoints are represented as a graph structure and processed using a **Pose Graph Convolutional Network (PGCN)** to learn spatial relationships between body joints.

To effectively integrate the information from both modalities, a **hierarchical cross-attention fusion mechanism** is employed. This fusion strategy enables the model to combine complementary features from facial expressions and body posture to produce a more reliable emotional representation. The proposed system is evaluated using the **RAF-DB (Real-world Affective Faces Database)** dataset, which contains real-world facial images categorized into seven basic emotion classes: angry, disgust, fear, happy, sad, surprise, and neutral.

Experimental results demonstrate that the proposed multimodal framework improves the robustness and accuracy of emotion recognition compared to traditional face-only methods. The system achieves an overall classification accuracy of approximately 84%, indicating that integrating facial and posture information provides a more comprehensive understanding of human emotional states.

CHAPTER - 1

INTRODUCTION

1.1 INTRODUCTION

Human emotions play a vital role in communication, decision-making, and social interaction. The ability to automatically recognize human emotions using computer systems has become an important research area in artificial intelligence and machine learning. Emotion recognition systems aim to identify and classify human emotional states based on various observable signals such as facial expressions, body posture, speech, and behavioral patterns.

Among these modalities, facial expressions have traditionally been the most widely used source of emotional information. With the advancement of deep learning techniques, convolutional neural networks (CNNs) have been extensively applied for facial emotion recognition due to their strong capability in learning visual features from images. However, CNN-based approaches mainly focus on localized spatial features and often face difficulties when the facial region is partially occluded, poorly illuminated, or captured at different viewing angles. These limitations reduce the reliability of emotion recognition systems in real-world environments.

In recent years, transformer-based models have gained significant attention in computer vision tasks. Vision Transformers (ViT) provide an alternative approach to traditional convolution-based models by using self-attention mechanisms to capture global relationships within an image. This allows the model to understand the overall facial structure and contextual information more effectively. As a result, transformer-based models have shown promising performance in visual recognition tasks, including facial expression analysis.

Apart from facial expressions, human body posture also provides valuable cues for understanding emotional states. For example, gestures such as raised arms, slumped shoulders, or head orientation can reflect different emotions. These posture-related features can be represented using skeletal keypoints detected from images. Graph-based deep learning models such as Graph Convolutional Networks (GCNs) are well suited for analyzing such structured data because they can learn spatial relationships between connected body joints.

Considering the complementary nature of facial expressions and body posture, combining multiple modalities can significantly improve the performance of emotion recognition systems. This project proposes a **multimodal emotion recognition framework** that integrates facial appearance and body posture information. Facial features are extracted using a **Vision Transformer**, while posture features are modeled using a **Pose Graph Convolutional Network (PGCN)**. A **hierarchical cross-attention fusion mechanism** is employed to combine both modalities and generate a unified emotional representation for classification.

The proposed approach aims to develop a more robust and reliable emotion recognition system that can perform effectively even under challenging real-world conditions such as occlusion, illumination variations, and pose changes. Emotion recognition has become an essential component in the development of intelligent systems that aim to interact naturally with humans. With the rapid growth of artificial intelligence, there is an increasing demand for systems that can understand not only human commands but also emotional states. This enables machines to respond more effectively and provide personalized experiences.

In real-world applications, emotions are often expressed through a combination of facial expressions, body language, and contextual cues. Relying on a single source of information may lead to incomplete or inaccurate interpretations. Therefore, integrating multiple modalities has become a key research direction in emotion recognition.

The advancement of deep learning techniques has significantly improved the capability of machines to analyze complex data. Models such as convolutional neural networks and transformers have enabled automatic feature extraction from large datasets. However, the challenge remains in developing systems that can perform reliably under diverse real-world conditions.

The proposed multimodal approach addresses these challenges by combining facial and posture information, thereby enhancing the robustness and accuracy of emotion recognition systems.

1.1.1 Applications of Emotion Recognition

Emotion recognition systems have gained significant importance in recent years due to their wide range of real-world applications. These systems enable machines to understand human emotional states and respond accordingly, improving interaction between humans and intelligent systems.

One of the major applications of emotion recognition is in **healthcare systems**. Emotion detection can help monitor the mental health of patients by identifying signs of stress, anxiety, or depression. It can also assist doctors in providing better treatment by understanding the emotional condition of patients.

Another important application is in **smart classrooms**. Emotion recognition systems can analyze students' engagement levels by detecting their emotions during lectures. This helps teachers understand whether students are attentive, confused, or bored, allowing them to improve teaching methods.

Emotion recognition is also widely used in **human–computer interaction (HCI)**. By understanding user emotions, systems can provide personalized responses, making interactions more natural and effective. For example, virtual assistants and chatbots can adapt their responses based on the user's emotional state.

In addition, emotion recognition plays a role in **intelligent surveillance systems**. It can help detect suspicious or unusual behavior by analyzing emotional expressions in crowded environments. This can improve security in public places such as airports, railway stations, and shopping malls.

Another emerging application is in the field of **entertainment and gaming**, where systems can adjust game difficulty or content based on the player's emotional reactions, enhancing user experience. Overall, emotion recognition systems have the potential to improve various domains by enabling machines to better understand and respond to human emotions.

1.1.2 Challenges in Emotion Recognition

Emotion recognition is a challenging task due to the complexity and variability of human emotions. Although significant progress has been made using deep learning techniques, several factors still affect the accuracy and reliability of emotion recognition systems.One of the major challenges is **facial occlusion**, where parts of the face are hidden due to objects such as masks, glasses, or hands. This makes it difficult for the model to capture important facial features required for accurate emotion classification.

Another challenge is **variation in lighting conditions**. Images captured under different lighting environments may affect the visibility of facial expressions, leading to incorrect predictions. Poor illumination or shadows can distort important facial features. **Pose variation** is also a significant issue. When the face is not directly facing the camera or is tilted at different angles, it becomes difficult for the system to analyze facial expressions effectively. In addition, **intra-class variation** and **inter-class similarity** make emotion recognition more complex. The same emotion may appear differently for different individuals, while different emotions may sometimes look similar. For example, fear and surprise often share similar facial expressions.

Another important challenge is **dataset imbalance**, where some emotion classes have more samples than others. This can cause the model to become biased toward frequently occurring emotions. Finally, real-world environments introduce noise, background variations, and unpredictable conditions, which further reduce the performance of emotion recognition systems. Addressing these challenges is essential for developing a robust and reliable emotion recognition system that performs effectively in practical scenarios.

1.2 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop a robust and efficient **multimodal emotion recognition system** capable of accurately identifying human emotions by analyzing both facial expressions and body posture using advanced deep learning techniques. The system aims to improve the reliability and accuracy of emotion recognition in real-world scenarios where facial information alone may be insufficient due to occlusion, lighting variations, or pose changes.

Below are the expanded goals of the project:

1. **To Develop a Multimodal Emotion Recognition Framework:**
   Design a system that integrates multiple sources of emotional information, particularly facial expressions and body posture, to create a more reliable emotion recognition model. By combining these modalities, the system can capture complementary emotional cues and improve classification performance.
2. **To Perform Effective Data Preprocessing:**
   Implement preprocessing techniques such as face detection, image resizing, normalization, and pose keypoint extraction to prepare the input data for deep learning models. Proper preprocessing helps in reducing noise and ensuring consistent input for accurate feature learning.
3. **To Extract Facial Features Using Vision Transformers (ViT):**
   Apply the Vision Transformer architecture to analyze facial images and capture global contextual relationships across facial regions. This approach enables the model to understand facial structures and micro-expressions more effectively compared to traditional convolutional neural networks.
4. **To Analyze Body Posture Using Pose Graph Convolutional Networks (PGCN):**
   Detect human pose keypoints using pose estimation techniques and represent them as graph structures where body joints act as nodes and their connections form edges. Use Pose Graph Convolutional Networks to learn spatial relationships between joints and extract posture-based emotional features.
5. **To Integrate Multimodal Features Using Hierarchical Cross-Attention Fusion:**
   Develop a fusion mechanism that combines facial features and posture features through a cross-attention strategy. This method allows the model to focus on the most relevant information from each modality and generate a unified emotional representation.
6. **To Improve Emotion Classification Accuracy:**
   Train the multimodal model using the RAF-DB dataset and evaluate its performance using metrics such as accuracy, precision, recall, and F1-score. The goal is to achieve higher classification accuracy compared to traditional single-modality emotion recognition systems.
7. **To Build a Reliable Emotion Prediction System:**
   Design a complete deep learning pipeline that processes input images, extracts multimodal features, performs classification, and outputs the predicted emotion category among the basic emotions such as Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

1.3 PROBLEM STATEMENT

Emotion recognition is an important area in artificial intelligence that enables machines to understand and interpret human emotional states. Accurate emotion detection can significantly improve applications such as human–computer interaction, healthcare monitoring, intelligent tutoring systems, and smart surveillance. However, developing a reliable emotion recognition system remains a challenging task due to the complexity and variability of human emotional expressions.

Many existing emotion recognition systems rely primarily on **facial expressions** for identifying emotions. Although facial features provide valuable information, systems based only on facial analysis often face difficulties in real-world scenarios. Factors such as **occlusion, lighting variations, head pose changes, and image quality** can reduce the visibility of facial features, leading to incorrect predictions.

In addition, traditional methods based on convolutional neural networks (CNNs) mainly focus on local spatial features of the face and may fail to capture global contextual relationships across different facial regions. This limitation affects the model's ability to understand complex emotional patterns. Human body posture also conveys important emotional cues. Gestures such as slouched shoulders, raised arms, or head orientation can reflect emotional states even when facial expressions are unclear. However, many conventional emotion recognition systems do not effectively utilize posture-based information.

Therefore, there is a need to develop a more robust and reliable emotion recognition framework that can analyze multiple sources of emotional information simultaneously. The problem addressed in this project is to design a multimodal emotion recognition system that integrates facial feature extraction using Vision Transformers and posture analysis using Pose Graph Convolutional Networks. By combining these modalities through an effective feature fusion mechanism, the system aims to improve emotion classification accuracy and provide more reliable predictions in real-world conditions.

CHAPTER - 2

LITERATURE REVIEW

Emotion recognition is an important research area in artificial intelligence that focuses on identifying human emotional states using computational techniques. Emotions play a crucial role in communication and social interaction, and the ability of machines to recognize emotions can significantly enhance applications such as human–computer interaction, healthcare monitoring, intelligent tutoring systems, surveillance systems, and customer behavior analysis. Over the past few decades, researchers have proposed several techniques for emotion recognition using different modalities such as facial expressions, speech signals, body gestures, and physiological signals. This chapter reviews the major approaches used in emotion recognition and highlights the contributions and limitations of previous research works. In recent years, the field of emotion recognition has witnessed significant advancements due to the integration of machine learning and deep learning techniques. Researchers have explored various approaches to improve the accuracy and reliability of emotion detection systems by utilizing different types of data and computational models.

The evolution of emotion recognition methods can be broadly categorized into traditional machine learning approaches, deep learning-based methods, and multimodal frameworks. Each of these approaches has contributed to the development of more advanced and efficient systems.

Furthermore, the availability of large-scale datasets and high-performance computing resources has enabled the training of complex deep learning models. This has led to improved generalization and better performance in real-world scenarios.

The study of emotion recognition continues to evolve with the introduction of advanced deep learning architectures and multimodal learning techniques. One of the key challenges in this domain is the ability to generalize across different environments and user conditions. Real-world datasets often contain noise, variations, and inconsistencies that can affect model performance.

To address these issues, researchers are increasingly focusing on robust feature extraction methods that can capture invariant patterns across different scenarios. Techniques such as attention mechanisms, transfer learning, and domain adaptation are being explored to improve model generalization.

Another important research direction is the development of lightweight models that can be deployed on edge devices such as mobile phones and embedded systems. While deep learning models provide high accuracy, their computational complexity often limits real-time deployment. Therefore, optimizing models for efficiency while maintaining performance is a key area of focus.

In addition, explainability in emotion recognition systems is gaining attention. Understanding how models make decisions is crucial for building trust in real-world applications. Visualization techniques and attention maps are used to interpret model predictions and identify important features contributing to emotion classification.

Overall, the literature indicates that combining multiple modalities with advanced deep learning architectures provides the most promising approach for achieving accurate and reliable emotion recognition.

2.1 Traditional Emotion Recognition Approaches

In the early stages of emotion recognition research, most systems relied on handcrafted feature extraction techniques combined with traditional machine learning classifiers. These methods focused on identifying distinctive visual patterns from facial images that could represent emotional expressions.

Commonly used handcrafted features included Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), Gabor filters, and Scale-Invariant Feature Transform (SIFT). These techniques extracted texture, edge, and gradient information from facial images. The extracted features were then used as inputs for classification algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Naïve Bayes, and Random Forest classifiers to categorize emotions.

Although these traditional approaches provided initial progress in emotion recognition research, they had several limitations. Handcrafted features required extensive domain knowledge and manual feature engineering, which made the system design complex. Moreover, these features often failed to capture subtle emotional variations and were highly sensitive to environmental factors such as illumination changes, head pose variations, and facial occlusions.

Another limitation of traditional approaches was their inability to automatically learn hierarchical features from large datasets. As a result, their performance was limited when applied to real-world scenarios containing diverse facial expressions and environmental conditions. These challenges motivated researchers to explore more advanced methods based on deep learning.

In addition to the commonly used techniques, traditional methods often rely on manually designed feature extraction processes. These handcrafted features require domain expertise and may not capture complex emotional patterns effectively. As a result, the performance of such systems is limited when dealing with real-world data that includes variations in lighting, pose, and facial expressions.

Another limitation of traditional approaches is their inability to generalize across different datasets. Since these methods are designed based on specific assumptions, they may fail when applied to new environments. This highlights the need for more advanced techniques that can automatically learn features from data.

2.2 Deep Learning Based Facial Emotion Recognition

The introduction of **deep learning techniques**, particularly Convolutional Neural Networks (CNNs), significantly improved the performance of emotion recognition systems. CNNs are capable of automatically learning hierarchical feature representations directly from raw image data without requiring manual feature extraction.Several deep learning architectures have been successfully applied for facial emotion recognition. Models such as **VGGNet, AlexNet, ResNet, and Xception** have demonstrated strong performance in image classification tasks. These networks consist of multiple convolutional layers that extract increasingly complex visual features from facial images. Early layers capture low-level features such as edges and textures, while deeper layers capture high-level features such as facial structures and expression patterns.

CNN-based emotion recognition systems have shown improved accuracy compared to traditional machine learning methods. They are capable of learning discriminative facial features and adapting to variations in facial expressions. Additionally, large-scale emotion datasets such as **RAF-DB, FER2013, and AffectNet** have enabled researchers to train deep learning models with better generalization capabilities. Despite these advantages, CNN-based models also have certain limitations. CNNs primarily focus on **local receptive fields**, meaning that they analyze small regions of the image at a time. This restricts their ability to capture long-range dependencies and global contextual relationships between distant facial regions.

As a result, CNN-based systems may fail to recognize emotions when important contextual information is distributed across different parts of the face. Another challenge is that CNN models may struggle in situations where facial features are partially hidden due to occlusion, accessories such as glasses or masks, or extreme head poses. These limitations encouraged researchers to investigate alternative architectures that could capture global relationships more effectively.

Deep learning methods overcome many limitations of traditional approaches by enabling automatic feature extraction. These models learn hierarchical representations of data, allowing them to capture both low-level and high-level features. This makes them highly effective for emotion recognition tasks.

The use of large datasets and powerful computational resources has further improved the performance of deep learning models. Techniques such as data augmentation and transfer learning help in enhancing model accuracy and generalization.

However, deep learning models also require careful tuning of parameters and sufficient training data to achieve optimal performance.

2.2.1 Architecture of Convolutional Neural Network

A Convolutional Neural Network (CNN) is a deep learning architecture widely used for image processing and computer vision tasks. CNNs are designed to automatically learn hierarchical feature representations from input images through multiple layers. The basic architecture of a CNN consists of several layers, including convolutional layers, pooling layers, activation functions, and fully connected layers. Each of these layers plays a specific role in extracting and processing features from images.

The convolutional layer applies filters to the input image to detect important features such as edges, textures, and patterns. These filters slide across the image and produce feature maps that highlight relevant information. Pooling layers are used to reduce the spatial dimensions of feature maps. This helps in reducing computational complexity and prevents overfitting. Common pooling operations include max pooling and average pooling.

Activation functions such as ReLU (Rectified Linear Unit) introduce non-linearity into the network, allowing it to learn complex patterns. Fully connected layers are used at the final stage to perform classification based on extracted features. CNNs have been widely used in facial emotion recognition due to their ability to automatically learn meaningful visual features from images.

2.2.2 Limitations of CNN in Emotion Recognition

Despite their effectiveness, CNN-based models have certain limitations in emotion recognition tasks. One of the main drawbacks is their focus on local receptive fields, which means they analyze only small regions of the image at a time.

This limitation makes it difficult for CNNs to capture global relationships between distant facial regions. Emotions often depend on the combination of multiple facial features, and ignoring global context may reduce classification accuracy. CNNs are also sensitive to variations in lighting conditions, pose, and occlusion. When important facial features are partially hidden or distorted, CNN models may fail to recognize emotions correctly.

These limitations highlight the need for advanced architectures such as Vision Transformers that can capture global dependencies more effectively. Deep learning has revolutionized the field of computer vision by enabling automatic feature extraction from large-scale datasets. Unlike traditional machine learning methods, deep learning models learn hierarchical representations of data through multiple layers of abstraction.

At the core of deep learning are artificial neural networks, which are inspired by the structure and functioning of the human brain. These networks consist of interconnected neurons organized in layers, where each neuron processes input data and passes the output to the next layer.

In image processing tasks, convolutional neural networks (CNNs) are widely used due to their ability to capture spatial features. However, recent advancements have introduced transformer-based architectures that overcome some of the limitations of CNNs by capturing global dependencies within the data.

The success of deep learning models depends on several factors, including dataset quality, model architecture, training strategies, and optimization techniques. Large datasets enable models to learn diverse patterns, while well-designed architectures improve feature extraction capabilities.

Optimization techniques such as gradient descent and its variants play a crucial role in training deep learning models. These techniques adjust model parameters iteratively to minimize the loss function and improve prediction accuracy.

Regularization techniques such as dropout and weight decay are used to prevent overfitting, ensuring that the model generalizes well to unseen data. Additionally, batch normalization helps stabilize training and improve convergence speed.

With continuous advancements in deep learning, new architectures and techniques are being developed to improve performance and efficiency. These innovations are driving the development of more intelligent and capable systems for various applications, including emotion recognition.

2.3 Transformer-Based Models for Emotion Recognition

Transformer-based architectures have recently emerged as powerful models for various computer vision tasks. Originally developed for natural language processing, transformers rely on **self-attention mechanisms** that allow the model to learn relationships between different elements of the input data.

The **Vision Transformer (ViT)** adapts the transformer architecture for image analysis. Instead of processing images using convolution operations, the Vision Transformer divides an image into smaller patches and treats each patch as an input token. These tokens are then processed through multiple transformer encoder layers that use self-attention mechanisms to analyze relationships between patches.

One of the key advantages of Vision Transformers is their ability to capture **global contextual information** across the entire image. Unlike CNNs, which process images locally, transformer models analyze all image regions simultaneously. This allows them to better understand spatial relationships between facial regions such as the eyes, mouth, eyebrows, and cheeks, which play important roles in expressing emotions.

Recent studies have demonstrated that Vision Transformers can outperform traditional CNN architectures in several visual recognition tasks. Their ability to capture global dependencies makes them particularly suitable for complex tasks such as facial emotion recognition.

However, most transformer-based emotion recognition systems still rely primarily on **facial information alone**. While facial expressions provide valuable emotional cues, they may not always be sufficient to accurately determine emotional states in real-world scenarios.

Another important advantage of transformer-based models is their ability to process input data in parallel, unlike sequential models. This parallel processing capability improves computational efficiency and reduces training time for large datasets.

Transformers also provide flexibility in handling different types of data, making them suitable for integration into multimodal systems. Their attention mechanisms allow the model to dynamically focus on important regions of the input, which is particularly useful in emotion recognition tasks where subtle variations play a crucial role.

Transformer-based models represent a significant advancement in deep learning. Their ability to capture long-range dependencies makes them highly suitable for tasks that require global context understanding.

In emotion recognition, transformers can analyze relationships between different facial regions simultaneously. This enables the model to understand subtle variations in expressions that may not be captured by traditional methods.

The flexibility and scalability of transformer models make them an important component in modern emotion recognition systems.

2.4 Multimodal Emotion Recognition

Human emotions are complex and are expressed through multiple behavioral signals rather than a single modality. For this reason, researchers have increasingly focused on multimodal emotion recognition systems that combine information from multiple sources such as facial expressions, body posture, speech signals, and textual data.

Multimodal systems aim to capture complementary emotional cues from different modalities and integrate them into a unified representation. This approach helps improve recognition accuracy and robustness, particularly in situations where one modality provides incomplete or ambiguous information.

Different fusion strategies have been proposed to combine multimodal features. These include early fusion, late fusion, and feature-level fusion techniques. In early fusion, features from different modalities are combined before model training. In late fusion, predictions from separate models are combined to produce the final result. Feature-level fusion integrates intermediate feature representations from different modalities within the network.

Recently, **attention-based fusion methods** have gained popularity in multimodal learning. Attention mechanisms allow the model to focus on the most informative features from each modality while ignoring irrelevant information. This dynamic feature weighting helps improve the effectiveness of multimodal systems.

Several research studies have explored multimodal emotion recognition approaches. For example, Fang et al. proposed an attention-based multimodal framework that integrates textual, audio, and visual features using a Bi-LSTM attention network. Their approach demonstrated improved emotion recognition performance by capturing contextual relationships across modalities. However, the model required large datasets and involved significant computational complexity.

Multimodal emotion recognition systems combine information from multiple sources to improve performance. By integrating facial expressions and posture data, these systems provide a more comprehensive understanding of emotional states.

The fusion of multiple modalities helps in reducing ambiguity and improving accuracy. It also allows the system to perform well under challenging conditions where one modality may be insufficient.

Recent research shows that multimodal approaches consistently outperform unimodal systems, making them a preferred choice for advanced emotion recognition applications.

2.5 Graph-Based Pose Analysis for Emotion Recognition

In addition to facial expressions, **human body posture and gestures** also provide important emotional cues. People often express emotions through body movements such as raising their arms when excited, lowering their shoulders when sad, or turning their head in response to certain emotional states.

To analyze body posture, researchers use **pose estimation techniques** that detect keypoints representing major body joints such as the shoulders, elbows, wrists, hips, and knees. These keypoints form a skeletal representation of the human body. This skeletal structure can be naturally represented as a **graph**, where body joints act as nodes and the physical connections between joints form edges. Graph-based representations allow researchers to analyze spatial relationships between different body parts.

Graph Convolutional Networks (GCNs) are specifically designed to process graph-structured data. They extend traditional convolution operations to graphs by aggregating information from neighboring nodes. This enables the network to learn spatial relationships between body joints and identify posture patterns associated with different emotions. Graph-based deep learning methods have been widely used in applications such as **human action recognition, gesture analysis, and behavior understanding**. However, only a limited number of studies have explored the integration of graph-based posture analysis with facial emotion recognition systems.

From the literature review, it is evident that emotion recognition has evolved significantly over time. Traditional methods laid the foundation, while deep learning techniques introduced automation and improved accuracy.

The emergence of transformer-based models and multimodal approaches has further enhanced system performance. These advancements highlight the importance of combining different techniques to achieve better results.

The proposed system builds upon these developments by integrating advanced architectures and multimodal feature fusion to improve emotion recognition accuracy.

2.6 Research Gap

From the analysis of existing literature, it is clear that significant progress has been made in emotion recognition using both traditional and deep learning techniques. CNN-based models have improved facial expression recognition, while transformer architectures have enhanced the ability to capture global image relationships. Additionally, graph-based models have shown strong potential for analyzing body posture and skeletal data.

Despite these advancements, many existing emotion recognition systems still rely primarily on **facial expressions alone**, which may not always provide sufficient information in real-world environments. Facial expressions can be affected by occlusion, lighting conditions, camera angles, and facial accessories, which may reduce recognition accuracy.

Furthermore, although multimodal systems have been proposed, many of them lack effective mechanisms for integrating information from different modalities. Some fusion methods simply combine features without considering the relationships between modalities, which limits their effectiveness.

Therefore, there is a need to develop a **robust multimodal emotion recognition framework** that effectively combines facial appearance and body posture information using advanced deep learning architectures. To address this research gap, the proposed project integrates

Vision Transformer-based facial feature extraction with **Pose Graph Convolutional Network-based posture analysis**. A **hierarchical cross-attention fusion mechanism** is employed to combine multimodal features and improve emotion classification performance under real-world conditions.

2.7 Comparative Analysis of Existing Methods

A comparative analysis of existing emotion recognition methods helps in understanding the strengths and limitations of different approaches. Various techniques have been proposed in the literature, ranging from traditional machine learning methods to advanced deep learning and multimodal frameworks.

Traditional methods based on handcrafted features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Support Vector Machines (SVM) provided initial solutions for emotion recognition. However, these methods were limited in their ability to capture complex emotional patterns and were highly sensitive to variations in lighting, pose, and occlusion.

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), improved emotion recognition performance by automatically learning hierarchical features from images. CNN-based models such as VGGNet, ResNet, and AlexNet demonstrated strong performance in facial emotion recognition tasks. However, CNNs primarily focus on local spatial features and may fail to capture global contextual relationships across different facial regions.

Transformer-based models, such as Vision Transformers (ViT), have further improved performance by using self-attention mechanisms to capture global dependencies in images. These models provide a better understanding of spatial relationships across the entire face and have shown promising results in emotion recognition tasks.

Multimodal emotion recognition systems combine multiple sources of information such as facial expressions, body posture, and speech signals. These systems provide improved accuracy and robustness compared to single-modality approaches. By integrating complementary emotional cues, multimodal systems can handle challenging real-world conditions more effectively.

The proposed system in this project combines Vision Transformers and graph-based pose analysis using a hierarchical cross-attention fusion mechanism. This approach addresses the limitations of previous methods by integrating both facial and posture information for improved emotion recognition performance.

Method	Technique Used	Advantages	Limitations
Traditional Methods	LBP, HOG, SVM	Simple, low computation	Low accuracy
CNN-based Methods	VGG, ResNet	Good feature extraction	Local features only
Transformer-based	Vision Transformer	Global understanding	Requires more data
Multimodal Methods	Face + Pose	High accuracy	Complex model

Table 2.1: Comparison of Emotion Recognition Methods

Emotion recognition is a challenging task due to the complexity and variability of human emotions. One of the major challenges is the presence of subtle differences between emotions. Expressions such as fear and surprise or sadness and neutral often appear similar, making classification difficult.

Another challenge is variation in lighting conditions. Poor lighting can obscure important facial features, reducing the effectiveness of feature extraction techniques. Similarly, occlusion caused by objects such as glasses, masks, or hair can affect model performance.

Pose variation is another important factor. Changes in head orientation or body posture can alter the appearance of facial features, making it difficult for the model to interpret emotions correctly.

Dataset imbalance is also a common issue in emotion recognition. Some emotion classes may have more samples than others, leading to biased learning. Techniques such as data augmentation and weighted loss functions are used to address this issue.

Real-world conditions introduce additional challenges such as background noise, image blur, and low resolution. These factors can affect the quality of input data and reduce model accuracy.

Despite these challenges, the proposed multimodal approach helps in improving performance by combining multiple sources of information.

CHAPTER - 3
PROPOSED METHOD

This chapter describes the design and development of the proposed multimodal emotion recognition system. The system integrates facial expression analysis and body posture analysis to improve the accuracy and robustness of emotion classification. Traditional emotion recognition systems mainly depend on facial expressions, which may become unreliable when facial information is partially occluded or affected by environmental conditions.

To overcome these limitations, the proposed framework utilizes multiple deep learning techniques. Facial features are extracted using a **Vision Transformer (ViT)** model that captures global contextual relationships in facial images. In addition to facial information, body posture features are obtained by extracting skeletal keypoints using **MediaPipe Pose** and modeling their spatial relationships using a **Pose Graph Convolutional Network (PGCN)**.

The extracted facial and posture features are then integrated using a **Hierarchical Cross-Attention Fusion mechanism**, which allows the model to effectively combine complementary information from both modalities. The fused features are passed to fully connected layers for final emotion classification into seven basic emotion categories.

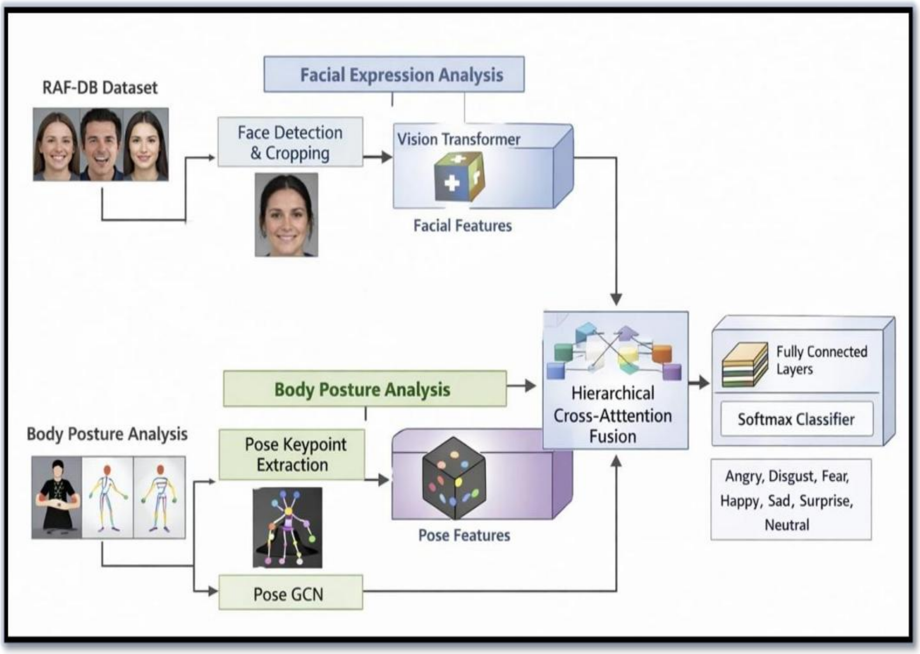

Figure 3 : System Architecture

The above figure provides a clear representation of the corresponding module in the proposed emotion recognition system. It illustrates how data flows through different stages and how various components interact with each other.

The visual representation helps in understanding the internal working of the system in a simplified manner. Each block shown in the figure represents a specific operation that contributes to the overall processing pipeline.

By analyzing this figure, it becomes easier to understand how input data is transformed into meaningful features and how these features are used for classification. The structured flow ensures that each stage performs its intended function efficiently.

Such diagrammatic representations are essential in deep learning systems, as they provide better clarity compared to textual explanations alone.

The overall workflow of the proposed system includes data preprocessing, facial feature extraction, pose feature extraction, multimodal feature fusion, and emotion classification.

Deep learning has emerged as one of the most powerful technologies in artificial intelligence, particularly in the field of computer vision. It enables machines to automatically learn complex patterns from large amounts of data without the need for manual feature extraction. This capability has significantly improved the performance of various applications such as image classification, object detection, speech recognition, and emotion recognition.

At the core of deep learning are artificial neural networks, which are inspired by the structure of the human brain. These networks consist of multiple layers of interconnected neurons that process input data and extract meaningful features through hierarchical transformations. Each layer of the network learns different levels of abstraction, starting from low-level features such as edges and textures to high-level features such as shapes and patterns.

One of the key advantages of deep learning is its ability to perform automatic feature extraction. In traditional machine learning methods, feature extraction required manual effort and domain expertise. However, deep learning models can learn relevant features directly from raw data, making them more efficient and scalable.

In image-based tasks, convolutional neural networks (CNNs) have been widely used due to their ability to capture spatial features. CNNs use convolutional filters to scan images and detect important visual patterns. These filters help the model identify features such as edges, corners, and textures that are essential for understanding image content.

Despite their success, CNNs have certain limitations, particularly in capturing global relationships within images. They primarily focus on local regions, which may result in the loss of contextual information. To overcome this limitation, transformer-based models have been introduced in computer vision.

Transformers use self-attention mechanisms to analyze relationships between different parts of the input data. In the context of images, Vision Transformers divide images into smaller patches and process them as sequences. This allows the model to capture global dependencies and understand the overall structure of the image more effectively.

Another important aspect of deep learning is model optimization. Training deep learning models involves minimizing a loss function using optimization algorithms such as gradient descent. The optimizer adjusts model parameters iteratively to reduce prediction error and improve accuracy.

The learning rate is a critical parameter in optimization. A high learning rate may cause the model to overshoot the optimal solution, while a low learning rate may slow down convergence. Therefore, selecting an appropriate learning rate is essential for effective training.

Regularization techniques are also important in deep learning. Overfitting occurs when a model performs well on training data but fails to generalize to new data. Techniques such as dropout, batch normalization, and weight decay help prevent overfitting and improve model generalization.

Dropout randomly disables a fraction of neurons during training, which prevents the model from relying too heavily on specific features. Batch normalization normalizes the inputs to each layer, improving training stability and convergence speed.

Another important concept in deep learning is transfer learning. Instead of training a model from scratch, transfer learning allows the use of pre-trained models that have already learned useful features from large datasets. This approach reduces training time and improves performance, especially when the available dataset is limited.

Data plays a crucial role in deep learning. The quality and diversity of the dataset directly affect model performance. Large and diverse datasets enable models to learn robust and generalized features. Data preprocessing and augmentation techniques further enhance dataset quality by introducing variations that improve model robustness.

In emotion recognition, combining multiple modalities has proven to be more effective than using a single modality. Human emotions are complex and are expressed through various signals such as facial expressions, body posture, and speech. By integrating multiple sources of information, multimodal systems can achieve higher accuracy and reliability.

Feature fusion is a critical component of multimodal systems. It involves combining features from different modalities to create a unified representation. Advanced fusion techniques such as attention-based methods allow the model to focus on the most relevant features from each modality.

The use of graph-based models for posture analysis is another significant advancement in emotion recognition. Graph Convolutional Networks are capable of processing structured data such as skeletal keypoints. These models capture spatial relationships between body joints and provide valuable information about posture-based emotional cues.

The combination of Vision Transformers and Graph Convolutional Networks represents a powerful approach for emotion recognition. While Vision Transformers capture global facial features, graph-based models analyze structural posture information. The integration of these modalities enhances the system's ability to recognize emotions accurately.

In practical applications, emotion recognition systems must operate under diverse and challenging conditions. Factors such as lighting variations, occlusion, and background noise can affect performance. Therefore, developing robust systems that can handle such conditions is essential.

Real-world deployment of emotion recognition systems requires efficient and scalable models. While deep learning models provide high accuracy, their computational complexity may limit real-time applications. Optimizing models for performance and efficiency is an important area of research.

Explainability is another important aspect of modern AI systems. Understanding how a model makes decisions is crucial for building trust and transparency. Visualization techniques such as attention maps can help interpret model predictions and identify important features.

The future of emotion recognition lies in the integration of advanced technologies such as multimodal learning, attention mechanisms, and real-time processing. These advancements will enable the development of more intelligent and human-aware systems.

Emotion recognition systems have the potential to transform various domains, including healthcare, education, entertainment, and security. By enabling machines to understand human emotions, these systems can improve user experience and facilitate more natural interactions.

Continuous research and development in this field will lead to more accurate, efficient, and reliable emotion recognition systems. The proposed multimodal approach represents a step forward in this direction by combining multiple sources of information and leveraging advanced deep learning techniques.

3.1 METHODOLOGY

The proposed emotion recognition system follows a **multimodal deep learning framework** that combines facial expression analysis and body posture analysis. The purpose of this approach is to improve emotion recognition accuracy by using complementary information from multiple modalities instead of relying only on facial expressions.

The workflow of the system consists of several stages including **data acquisition, preprocessing, facial feature extraction, pose feature extraction, feature fusion, and emotion classification**. Initially, images are obtained from the RAF-DB dataset, which contains real-world facial images with different emotional expressions.

During the preprocessing stage, face detection is performed to locate and crop facial regions from the input images. The cropped faces are resized to a fixed resolution and normalized to ensure consistent input for the deep learning model. At the same time, human pose keypoints are extracted using MediaPipe Pose, which detects important body joints such as shoulders, elbows, wrists, and hips.

The system then processes the input through two parallel branches. In the first branch, facial images are analyzed using a **Vision Transformer (ViT)** model to capture global facial patterns. In the second branch, body posture information is modeled using a **Pose Graph Convolutional Network (PGCN)** that learns spatial relationships between body joints.

The extracted facial and posture features are then combined using a **hierarchical cross-attention fusion mechanism**. This fusion strategy allows the model to integrate complementary information from both modalities to generate a unified feature representation.

Finally, the fused features are passed to fully connected layers followed by a **Softmax classifier**, which predicts the final emotion category among seven classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

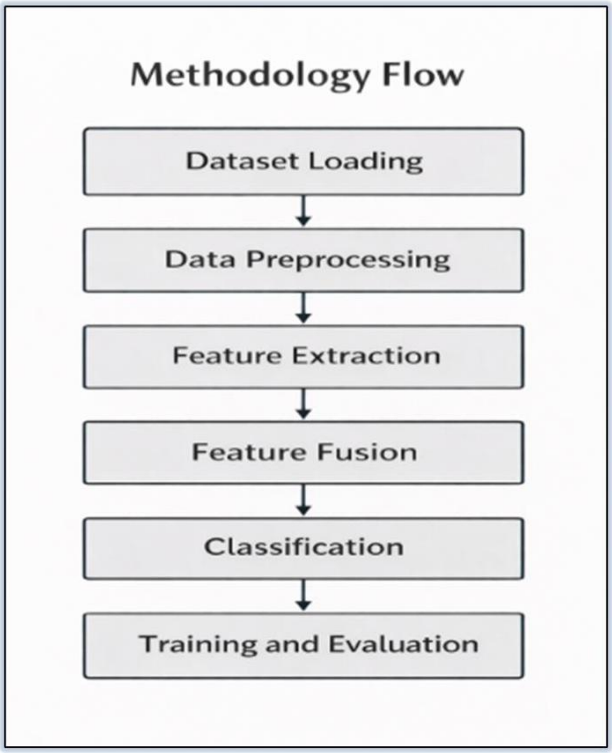

Figure 3.1: Methodology

The working of the proposed system can be further explained in a step-by-step manner to provide a clear understanding of each stage involved in emotion recognition.

Step 1: The input image is provided to the system from the dataset or real-time source.

Step 2: The image undergoes preprocessing to remove noise and improve quality.

Step 3: Face detection is performed to isolate the facial region.

Step 4: Pose keypoints are extracted to capture body posture information.

Step 5: Facial features are extracted using the Vision Transformer model.

Step 6: Posture features are extracted using graph-based techniques.

Step 7: Both feature sets are combined using a fusion mechanism.

Step 8: The fused features are passed to the classification layer.

Step 9: The final emotion is predicted using the Softmax function.

Step 10: The output is displayed along with confidence values.

This step-by-step explanation provides a clear view of how the system processes input data and generates output predictions.

The methodology diagram illustrates the overall workflow of the proposed multimodal emotion recognition system. The system processes input images through several stages including image preprocessing, facial feature extraction, pose keypoint detection, multimodal feature fusion, and emotion classification.

Initially, the input image is preprocessed to detect and crop the facial region while also extracting pose keypoints representing body joints. The facial image is then processed using a **Vision Transformer (ViT)** to extract facial features, while the pose keypoints are analyzed using a **Pose Graph Convolutional Network (PGCN)** to obtain posture-based features.

These features are combined through a **cross-attention fusion mechanism**, which integrates information from both modalities. The fused features are finally passed to a classification layer that predicts the emotion category among the seven basic emotions.

The proposed methodology is designed to ensure that each stage of the system contributes effectively to the overall performance. By combining preprocessing, feature extraction, and fusion techniques, the system achieves a balance between accuracy and computational efficiency.

The use of parallel processing for facial and posture analysis allows the system to capture complementary features simultaneously. This improves the model’s ability to recognize emotions even when one modality is less informative.

Additionally, the modular design of the system makes it flexible and scalable. Each component can be improved or replaced independently, allowing future enhancements without redesigning the entire system.

3.1.1 Working of the Proposed System

The proposed multimodal emotion recognition system follows a structured pipeline that processes input images through multiple stages to accurately predict human emotions. This section provides a step-by-step explanation of how the system works in a practical scenario.

In the first step, an input image is provided to the system either through a dataset or a user interface. The image may contain variations in lighting, pose, and background conditions. Therefore, preprocessing is performed to improve the quality and consistency of the input data.

During preprocessing, the system performs face detection to identify and crop the facial region from the input image. The cropped face is then resized to a fixed dimension suitable for the Vision Transformer model. At the same time, the system extracts pose keypoints using the MediaPipe Pose framework, which detects important body joints such as shoulders, elbows, and wrists.

In the next stage, the system processes the input through two parallel branches. The first branch focuses on facial feature extraction using the Vision Transformer. The input image is divided into patches, and each patch is converted into an embedding. These embeddings are passed through transformer layers, where self-attention mechanisms capture relationships between different facial regions.

The second branch focuses on posture analysis. The extracted pose keypoints are represented as a graph structure where joints act as nodes and connections between them act as edges. This graph is processed using a Pose Graph Convolutional Network (PGCN), which learns spatial relationships between body joints and extracts posture-based features.

After feature extraction, the system combines facial and posture features using a hierarchical cross-attention fusion mechanism. This allows the model to focus on the most relevant features from both modalities and generate a unified feature representation.

Finally, the fused feature vector is passed to fully connected layers, followed by a Softmax classifier, which predicts the final emotion category. The output is one of the seven emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, or Neutral.

This step-by-step workflow demonstrates how the proposed system integrates multiple modalities to achieve accurate and robust emotion recognition.

3.1.2 Algorithm of the Proposed System

The overall working of the proposed emotion recognition system can be summarized in the following steps:

Step 1: Load the RAF-DB dataset or input image
Step 2: Perform image preprocessing (face detection, resizing, normalization)
Step 3: Extract pose keypoints using MediaPipe Pose
Step 4: Extract facial features using Vision Transformer (ViT)
Step 5: Construct pose graph and extract posture features using PGCN
Step 6: Apply cross-attention fusion to combine features
Step 7: Pass fused features to classification layer
Step 8: Predict emotion using Softmax function
Step 9: Display final emotion output

3.2 DATASET DESCRIPTION

In this project, the performance of the proposed emotion recognition system is evaluated using the **RAF-DB (Real-world Affective Faces Database)** dataset. RAF-DB is a widely used benchmark dataset for facial expression recognition that contains images collected from real-world environments. The dataset includes facial images with different emotional expressions captured under various conditions such as changes in illumination, head pose, occlusion, and background variations.

RAF-DB consists of thousands of facial images annotated with emotion labels by human annotators. These annotations help in training deep learning models to accurately classify emotional states. The dataset is particularly suitable for emotion recognition research because it represents real-world scenarios rather than controlled laboratory conditions.

In this project, the **basic emotion subset** of RAF-DB is used. This subset contains images categorized into seven fundamental emotion classes that represent common human emotional states. These emotion categories include:

- Angry
- Disgust
- Fear
- Happy
- Sad

- Surprise
- Neutral

The dataset is typically divided into **training and testing sets**. The training set is used to train the deep learning model so that it can learn patterns associated with different emotions. The testing set is used to evaluate the performance of the trained model and measure how accurately it can classify unseen images.

Each emotion category in the dataset is organized into separate folders, allowing the system to easily load and process images during training and testing. The RAF-DB dataset provides a diverse set of facial expressions, making it suitable for developing robust emotion recognition systems.

Another important aspect of dataset utilization is the preprocessing pipeline applied before model training. The quality of input data directly influences the performance of deep learning models. Therefore, careful handling of dataset variations such as noise, blur, and background clutter is essential.

Data balancing techniques may also be applied to address class imbalance issues. Techniques such as oversampling, undersampling, or weighted loss functions can be used to ensure that all emotion classes are equally represented during training.

Additionally, splitting the dataset into training, validation, and testing subsets ensures that the model is evaluated fairly. The validation set is particularly useful for tuning hyperparameters and preventing overfitting during training. Proper dataset management and preprocessing contribute significantly to the success of the proposed emotion recognition system.

3.2.1 Dataset Distribution and Details

The RAF-DB dataset used in this project contains a large number of real-world facial images collected from various sources. These images are annotated with emotion labels and include significant variations in facial expressions, lighting conditions, background, and head pose. The dataset consists of thousands of images categorized into seven basic emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. Each class contains a different number of samples, which may lead to slight imbalance in the dataset distribution.

The distribution of images across different emotion categories plays an important role in training the model. A balanced dataset helps the model learn all emotion classes effectively, while an imbalanced dataset may cause the model to become biased toward certain classes. The dataset is divided into training and testing sets. The training set is used to train the deep learning model, while the testing set is used to evaluate its performance on unseen data. This separation ensures that the model generalizes well to new inputs.

Emotion	Approx. Number of Images
Angry	1200
Disgust	500
Fear	1000
Happy	3000
Sad	1500
Surprise	1200
Neutral	2500

Table 3.1: Sample Distribution of RAF-DB Dataset

3.2.2 Advantages of RAF-DB Dataset

The RAF-DB dataset offers several advantages for emotion recognition research. One of the key strengths of the dataset is that it contains images collected from real-world environments rather than controlled laboratory settings. This makes the dataset more suitable for developing robust emotion recognition systems.

Another advantage is the diversity of the dataset. The images include variations in age, gender, ethnicity, lighting conditions, and facial orientations. This diversity helps the model learn generalized features and improves its ability to perform well on unseen data.

The dataset also provides manually annotated emotion labels, ensuring high-quality ground truth data for training and evaluation. This improves the reliability of the model's predictions.

Furthermore, RAF-DB is widely used in research, making it easier to compare the performance of the proposed system with existing methods.

3.3 DATA PREPARATION

Before training the deep learning model, the dataset must be processed and organized to ensure that the input data is suitable for the proposed multimodal framework. Data preparation plays an important role in improving model performance by reducing noise and standardizing the input format.

In this project, the images from the RAF-DB dataset undergo several preprocessing operations before being used for emotion recognition. These operations include **face detection, image resizing, normalization, and pose keypoint extraction**. The purpose of these steps is to ensure that the model receives consistent and meaningful input data for both facial and posture analysis.

The facial images are first processed to detect and isolate the face region from the input image. After detecting the face, the cropped facial region is resized to a fixed resolution to match the input requirements of the Vision Transformer model. Pixel values are then normalized to improve training stability and accelerate the convergence of the deep learning model.

In addition to facial preprocessing, body posture information is extracted from each image using pose estimation techniques. The MediaPipe Pose framework is used to detect human pose keypoints corresponding to important body joints. These keypoints are later used to represent the human skeleton structure for posture-based emotion analysis.

The prepared facial images and pose keypoints are then used as synchronized multimodal inputs for the proposed emotion recognition system.

The data preprocessing stage plays a crucial role in improving the performance of the proposed system. Raw input images often contain noise, variations in lighting, and background clutter, which can negatively affect model performance.

To address these issues, multiple preprocessing steps are applied. Face detection is used to isolate the facial region from the input image, ensuring that the model focuses only on relevant features. The detected face is then resized to a fixed dimension suitable for the Vision Transformer.

Normalization is performed to scale pixel values to a standard range, which improves training stability. In addition, pose keypoints are extracted using MediaPipe, providing structural information about body posture.

Data cleaning techniques are also applied to remove corrupted or low-quality images. This ensures that the dataset used for training is consistent and reliable. These preprocessing steps collectively enhance the quality of input data and significantly improve the performance of the emotion recognition system.

Data preparation is one of the most critical stages in the development of a deep learning model. The quality of input data directly influences the performance and accuracy of the system. In emotion recognition, preprocessing becomes even more important due to variations in facial expressions, lighting conditions, and background environments.

The preprocessing pipeline ensures that the input data is consistent and suitable for model training. Techniques such as normalization, resizing, and noise reduction are applied to improve data quality. These steps help in reducing unwanted variations and allow the model to focus on relevant features.

Face detection plays a key role in isolating the region of interest from the input image. By focusing only on the facial region, the system avoids unnecessary background information that may affect performance. Similarly, pose keypoint extraction provides structural information about body posture, which complements facial features.

Data augmentation techniques further enhance the dataset by introducing variations such as rotation, flipping, and brightness adjustment. This helps the model generalize better and perform well on unseen data.

Proper data preparation ensures that the model receives high-quality input, leading to improved accuracy and robustness.

3.3.1 Image Preprocessing

Before extracting features for emotion recognition, the input images must undergo several preprocessing operations to ensure consistent and high-quality data for the deep learning model. Image preprocessing helps remove unnecessary background information, standardize image dimensions, and improve the stability of the training process.

The first step in preprocessing is **face detection**. Since the RAF-DB dataset contains real-world images with varying backgrounds, it is necessary to locate the facial region in each image. Face detection algorithms are used to identify the position of the face and crop the relevant facial area from the original image. This ensures that the model focuses only on the facial features that contribute to emotion recognition.

After detecting the face region, the cropped facial images are **resized to a fixed resolution** so that they can be processed by the Vision Transformer model. Standardizing the image size is important because deep learning models require consistent input dimensions during training and inference.

The final step in preprocessing is **pixel normalization**. The pixel values of the images are scaled to a normalized range, which helps improve numerical stability during model training. Normalization also allows the learning algorithm to converge faster and prevents large variations in pixel intensity from affecting the model's performance. These preprocessing steps ensure that the input images are clean, consistent, and suitable for feature extraction using deep learning techniques.

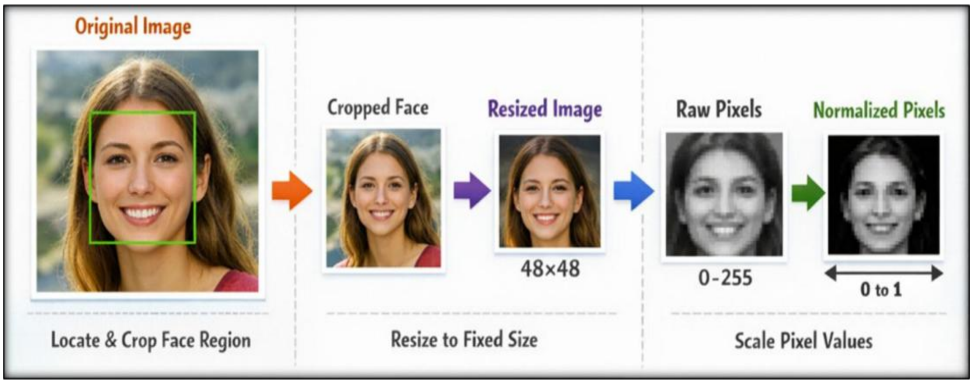

Figure 3.3.1 : Image preprocessing Pipeline

3.3.2 Pose Keypoint Extraction

In addition to facial information, body posture provides valuable cues for understanding human emotions. Certain emotions can be expressed not only through facial expressions but also through body gestures such as head orientation, shoulder position, and arm movements. Therefore, pose-based information is incorporated into the proposed multimodal emotion recognition system.

In this project, **MediaPipe Pose** is used to detect human body keypoints from input images. MediaPipe Pose is a real-time pose estimation framework developed by Google that can accurately identify major body joints from images or video frames. The framework detects several keypoints representing important body parts such as the shoulders, elbows, wrists, hips, knees, and ankles.

Each detected keypoint is represented by spatial coordinates that describe its position within the image. These keypoints collectively form a skeletal representation of the human body. The extracted keypoints are then normalized to reduce variations caused by differences in image scale and body orientation.

The normalized pose keypoints are later used to construct a graph structure where body joints act as nodes and the connections between joints represent edges. This graph representation allows the system to analyze spatial relationships between body joints and extract posture-based emotional features using graph convolutional networks.

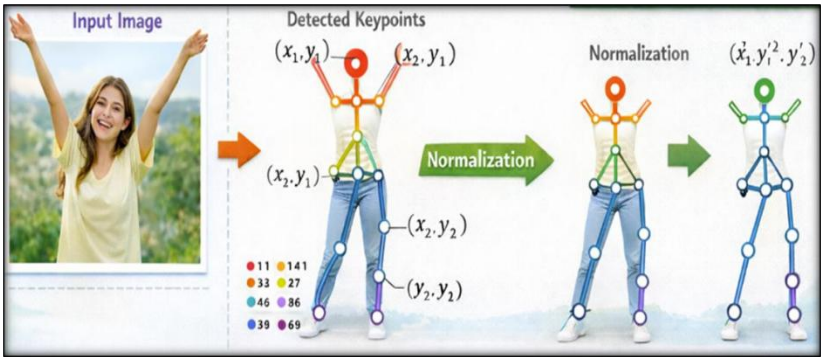

Figure 3.3.2 : Human pose keypoints detection using MediaPipe Pose

3.3.3 Data Augmentation Techniques

Data augmentation is an important technique used to improve the performance and generalization ability of deep learning models. It involves creating modified versions of existing images to increase the diversity of the training dataset without collecting new data.

In this project, data augmentation techniques are applied to enhance the robustness of the emotion recognition system. These techniques help the model learn variations in facial expressions and improve its ability to handle real-world conditions.

One common augmentation technique is **image rotation**, where images are slightly rotated at different angles. This helps the model become invariant to head tilts and orientation changes.

Another technique is **horizontal flipping**, where images are flipped along the vertical axis. This increases the number of training samples and allows the model to learn symmetrical facial patterns.

Scaling and resizing are also used to simulate variations in image size and distance from the camera. This helps the model handle images captured at different distances.

Additionally, **brightness and contrast adjustment** are applied to simulate different lighting conditions. This improves the model's performance under varying illumination environments.

Data augmentation helps reduce overfitting by exposing the model to diverse training samples, thereby improving its generalization capability.

3.3.4 Importance of Data Preparation

Data preparation is a crucial step in any machine learning or deep learning project, as it directly affects the performance of the model. Properly prepared data ensures that the model receives clean, consistent, and meaningful input for training.

In emotion recognition systems, preprocessing helps remove noise and irrelevant background information, allowing the model to focus on important facial and posture features. Techniques such as face detection, normalization, and pose keypoint extraction play a vital role in improving data quality.

Well-prepared data also helps in faster convergence during training and reduces the chances of model overfitting. It ensures that the model can learn meaningful patterns and generalize effectively to new and unseen data.

Therefore, careful data preparation is essential for building a robust and reliable emotion recognition system.

3.4 FACIAL FEATURE EXTRACTION USING VISION TRANSFORMER (ViT)

Facial feature extraction is an important step in the proposed emotion recognition system. In this project, a **Vision Transformer (ViT)** model is used to extract meaningful facial features from input images. Vision Transformers have recently gained popularity in computer vision tasks because they are capable of capturing global contextual relationships across an image using a self-attention mechanism.

Unlike traditional Convolutional Neural Networks (CNNs), which analyze local regions of an image using convolution filters, the Vision Transformer processes the entire image by dividing it into **fixed-size patches**. Each image is first split into small patches of equal size. These patches are then flattened and converted into embedding vectors through a linear projection layer. In addition to patch embeddings, **positional encoding** is added to preserve spatial information about the location of each patch within the image.

The embedded patches are then passed through a stack of **transformer encoder layers**. Each encoder layer contains two important components: **multi-head self-attention** and **feed-forward neural networks**. The self-attention mechanism allows the model to learn relationships between different image patches, enabling it to capture global dependencies between facial regions such as the eyes, mouth, and eyebrows.

Through multiple transformer layers, the model learns high-level facial representations that describe important features related to emotional expressions. The final output of the Vision

Transformer is a **high-dimensional feature vector** that summarizes the facial characteristics of the input image.

These extracted facial features are then used in the later stages of the system, where they are combined with posture features using a multimodal fusion mechanism to improve emotion recognition accuracy.

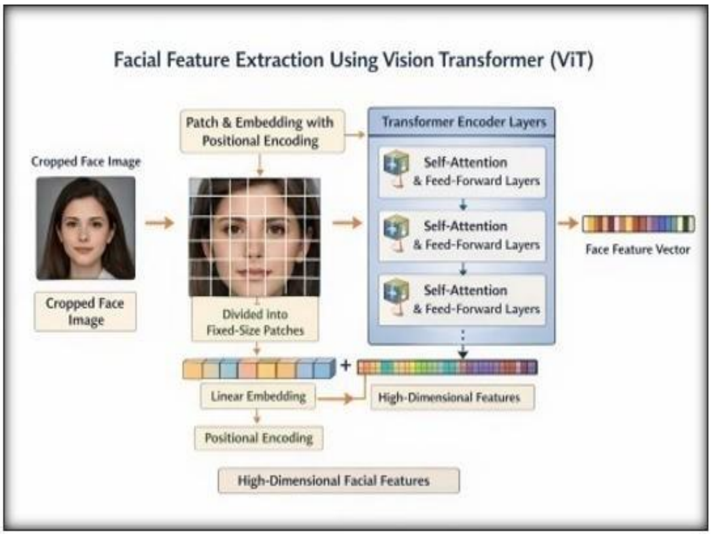

Figure 3.4 : Facial Feature Extraction using Vision Transformer (ViT)

The above figure illustrates the working architecture of the Vision Transformer used for facial feature extraction in the proposed system. The input image is first divided into smaller patches of equal size, which allows the model to process the image in segments rather than as a whole. Each patch is then flattened and converted into a vector representation through a linear embedding layer.

To retain the spatial information of the image, positional encoding is added to each patch embedding. This ensures that the model understands the relative position of each patch within the image. The sequence of embedded patches is then passed through multiple transformer encoder layers.

Each transformer encoder layer consists of multi-head self-attention and feed-forward neural network components. The self-attention mechanism allows the model to analyze relationships between different patches, enabling it to capture global dependencies across the image.

The Vision Transformer represents a significant advancement in the field of computer vision. Unlike traditional convolutional neural networks, it processes images by dividing them into patches and analyzing relationships between these patches using attention mechanisms.

One of the key advantages of Vision Transformers is their ability to capture global context. This is particularly useful in emotion recognition, where multiple facial features must be analyzed together to understand expressions.

The self-attention mechanism allows the model to focus on important regions of the image, improving feature extraction. This enables the system to identify subtle variations in facial expressions that may not be captured by conventional methods.

Despite its advantages, the Vision Transformer requires a large amount of data for training. However, when combined with proper preprocessing and training techniques, it provides excellent performance in emotion recognition tasks.

Through this process, the Vision Transformer learns high-level representations of facial features such as eyes, mouth, and eyebrows. These features play a crucial role in emotion recognition. The output of the transformer is a feature vector that summarizes the important characteristics of the facial image.

This approach improves the model's ability to understand complex facial expressions compared to traditional convolutional methods.

3.4.1 Working of Vision Transformer

The Vision Transformer (ViT) is a deep learning architecture that applies the transformer model, originally designed for natural language processing, to image classification tasks. Unlike traditional convolutional neural networks, Vision Transformers process images by dividing them into smaller patches and analyzing relationships between these patches using attention mechanisms.

In the first step, the input image is divided into fixed-size patches, typically of size 16×16 pixels. Each patch is then flattened into a one-dimensional vector and passed through a linear projection layer to generate patch embeddings. These embeddings represent different regions of the image.

Since transformers do not inherently capture positional information, positional encoding is added to each patch embedding. This helps the model understand the spatial arrangement of patches within the image.

The embedded patches are then passed through multiple transformer encoder layers. Each encoder layer consists of multi-head self-attention and feed-forward neural network components. The self-attention mechanism allows the model to analyze relationships between different patches, enabling it to capture global dependencies across the entire image.

Through this process, the Vision Transformer learns meaningful representations of facial features such as eyes, mouth, and eyebrows, which are essential for emotion recognition. The final output of the transformer is a feature vector that summarizes the important characteristics of the input image.

This component plays an important role in the overall functioning of the proposed system. It ensures that relevant features are effectively extracted and processed, which directly impacts the performance of the model. By handling variations in input data, this stage improves the robustness and reliability of the system.

3.4.2 Advantages of Vision Transformer

The Vision Transformer offers several advantages over traditional convolutional neural networks in image analysis tasks. One of the primary advantages is its ability to capture global relationships across the entire image. Unlike CNNs, which focus on local regions, Vision Transformers analyze all image patches simultaneously, allowing them to understand the overall context more effectively.

Another advantage is the use of self-attention mechanisms, which enable the model to focus on the most relevant parts of the image. This improves feature extraction and helps in identifying subtle facial expressions that are important for emotion recognition.

Vision Transformers are also highly scalable and can achieve better performance when trained on large datasets. They have shown state-of-the-art results in various computer vision tasks, including image classification and object detection.

Additionally, Vision Transformers provide flexibility in model design and can be easily integrated with other deep learning architectures, making them suitable for multimodal systems.

3.4.3 Limitations of Vision Transformer

Despite their advantages, Vision Transformers also have certain limitations. One of the main challenges is that they require a large amount of training data to perform effectively. Unlike CNNs, which have built-in inductive biases such as locality and translation invariance, transformers rely heavily on data to learn these patterns.

Another limitation is the high computational cost associated with self-attention mechanisms. Processing all image patches simultaneously requires significant memory and computational resources, which may not be suitable for low-power devices.

Vision Transformers may also struggle when trained on smaller datasets, as they may not generalize well without sufficient data. This makes it important to use appropriate datasets and training strategies to achieve optimal performance.

3.4.4 Practical Working of Vision Transformer (ViT)

The Vision Transformer (ViT) is used in the proposed system to extract meaningful facial features from input images. In this approach, the image is not processed as a whole; instead, it is divided into smaller patches and analyzed step by step.

Step 1: Input Image

The input image is represented as:

$X \in R^{(H \times W \times C)}$

Where:
H = height of the image
W = width of the image
C = number of channels (RGB = 3)

Example:
$X \in R^{(224 \times 224 \times 3)}$

Step 2: Image Patch Splitting

The image is divided into smaller patches of size $P \times P$.

Number of patches:

$N = (H \times W) / (P \times P)$

Example:
If patch size P = 16, then:

$N = (224 \times 224) / (16 \times 16) = 196$ patches

Each patch is flattened into a vector:

$xp \in R^{(P^2 \times C)}$

Step 3: Linear Patch Embedding

Each patch is converted into an embedding using a linear transformation:

$z0 = [xp1E; xp2E; ...; xpNE]$

Where:
E = embedding matrix

This step converts image patches into numerical vectors that the model can process.

Step 4: Positional Encoding

To maintain spatial information, positional encoding is added:

$z0 = z0 + Epos$

This helps the model understand the arrangement of facial features such as eyes, nose, and mouth.

Step 5: Self-Attention Mechanism

For each patch, three matrices are calculated:

$Q = XWQ$
$K = XWK$
$V = XWV$

Attention is computed as:

$\text{Attention}(Q, K, V) = \text{softmax}((Q \times K^T) / \sqrt{dk}) \times V$

This step allows the model to understand relationships between different parts of the face.

Step 6: Multi-Head Attention

Multi-head attention is applied as:

$MHA(X) = \text{Concat}(head1, head2, ..., headh)WO$

Each head focuses on different facial regions such as eyes, mouth, and eyebrows.

Step 7: Feed Forward Network

The output is refined using:

$$FFN(x) = \max(0, xW1 + b1)W2 + b2$$

This step improves feature representation.

Step 8: Final Feature Representation

The transformer generates a final feature vector:

$$zL = Transformer(z0)$$

This vector represents the complete facial expression.

Step 9: Emotion Classification

The final prediction is obtained using:

$$y = softmax(WzL + b)$$

Example output:

[Happy = 0.7, Sad = 0.2, Angry = 0.1]

The emotion with the highest probability is selected as the final output.

In the proposed system, the Vision Transformer processes the input image by dividing it into patches, converting them into embeddings, and applying attention mechanisms to understand relationships between facial features. The extracted features are then combined with posture features and passed to the classification layer to predict the final emotion.

3.5 POSE FEATURE EXTRACTION USING POSE GRAPH CONVOLUTIONAL NETWORK (PGCN)

In addition to facial expressions, human body posture also provides important cues for understanding emotional states. Certain emotions can be expressed through body gestures such as shoulder movement, head orientation, and arm positions. Therefore, posture information is incorporated into the proposed multimodal emotion recognition framework.

The first step in posture analysis is **pose keypoint detection**. In this project, the

MediaPipe Pose framework is used to detect human body keypoints from input images. MediaPipe Pose is a lightweight and efficient pose estimation model capable of identifying major body joints such as the shoulders, elbows, wrists, hips, knees, and ankles.

Each detected joint is represented as a **keypoint with spatial coordinates** indicating its position in the image. These keypoints collectively form a skeletal representation of the human body. The detected keypoints are then normalized to reduce variations caused by different image scales or camera distances.

To analyze relationships between body joints, the pose keypoints are represented as a **graph structure**. In this graph representation, each body joint is treated as a **node**, while the anatomical connections between joints form **edges**. This graph structure enables the model to capture spatial dependencies between different body parts.

A **Pose Graph Convolutional Network (PGCN)** is used to process this skeletal graph. Graph convolution operations allow the network to aggregate information from neighboring joints and learn structural patterns in body posture. Through multiple graph convolution layers, the PGCN extracts meaningful posture features that are useful for identifying emotional cues.

The resulting posture feature vector represents the structural body posture information of the person in the image. These features are later combined with facial features through a multimodal fusion mechanism to improve emotion classification performance.

Graph-based models provide an effective way to analyze structured data such as human body posture. By representing body joints as nodes and their connections as edges, the system can model spatial relationships between different parts of the body.

The use of Graph Convolutional Networks enables the system to learn patterns from this structured data. Each node updates its representation by aggregating information from neighboring nodes, allowing the model to capture interactions between body joints.

This approach is particularly useful in emotion recognition, as body posture often provides additional context that complements facial expressions. For example, a slouched posture may indicate sadness, while an upright posture may indicate confidence or happiness.

Graph-based models are robust to variations in appearance and can handle different body orientations effectively. This makes them suitable for real-world applications where input conditions may vary.

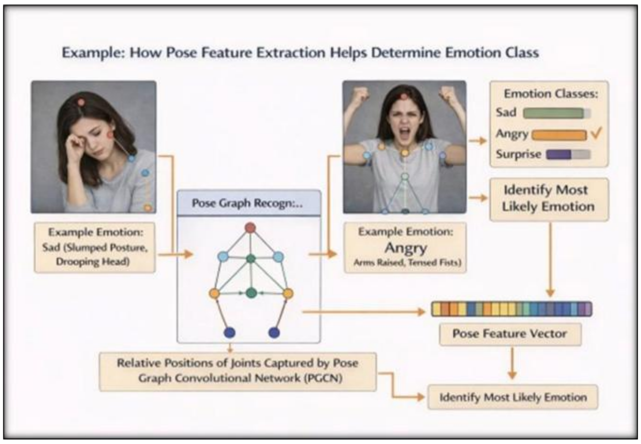

Figure 3.5 : Pose keypoints and graph representation

3.5.1 Graph Representation of Human Body

In the proposed emotion recognition system, the human body is represented as a graph structure to effectively analyze posture information. A graph is a mathematical structure consisting of nodes and edges. In this context, each body joint detected using pose estimation is treated as a node, while the connections between joints represent edges.

The pose keypoints extracted using MediaPipe Pose include important body joints such as shoulders, elbows, wrists, hips, knees, and ankles. These keypoints collectively form a skeletal representation of the human body.

By representing the human body as a graph, it becomes possible to model spatial relationships between different joints. For example, the relative position of the shoulder with respect to the elbow or the orientation of the head with respect to the torso can provide valuable information about emotional states.

This graph-based representation allows the system to capture structural information about body posture, which is not possible using traditional image-based approaches alone.

The effectiveness of this stage lies in its ability to process complex data and extract meaningful information. It contributes significantly to improving the accuracy of emotion recognition by ensuring that important patterns are captured and utilized during classification.

3.5.2 Working of Graph Convolutional Network (GCN)

Graph Convolutional Networks (GCNs) are a type of deep learning model designed to process graph-structured data. Unlike traditional neural networks that operate on grid-like data such as images, GCNs can learn patterns from data represented as graphs.

In a GCN, each node in the graph updates its feature representation by aggregating information from its neighboring nodes. This process is known as message passing. During each layer of the network, a node collects information from its connected nodes, combines it with its own features, and updates its representation.

Mathematically, this operation involves multiplying the adjacency matrix of the graph with the feature matrix, followed by a transformation using learnable weights. This allows the network to learn spatial dependencies between connected nodes.

In the context of this project, GCNs are used to analyze relationships between body joints. By learning how different joints move and interact, the model can identify posture patterns associated with different emotions.

This module enhances the system's capability to generalize across different input conditions. By addressing challenges such as noise and variations in data, it ensures consistent performance and improves the overall efficiency of the model.

3.5.3 Advantages of PGCN

The Pose Graph Convolutional Network (PGCN) provides several advantages for emotion recognition tasks. One of the main benefits is its ability to capture spatial relationships between body joints. Unlike traditional methods that treat keypoints independently, PGCN considers the structural connections between joints.

Another advantage is its effectiveness in modeling human posture. By analyzing the relative positions of body parts, the network can detect subtle posture changes that indicate emotional states.

PGCN is also robust to variations in image appearance, as it relies on skeletal representations rather than raw pixel values. This makes it less sensitive to changes in lighting, background, and image quality.

Additionally, graph-based models are well suited for handling structured data, making them an ideal choice for pose-based emotion analysis.

3.5.4 Practical Working of Pose Graph Convolutional Network (PGCN)

The Pose Graph Convolutional Network (PGCN) is used in the proposed system to extract posture-based features from human body keypoints. It represents the human body as a graph structure, where joints are treated as nodes and connections between them are treated as edges.

Step 1: Input Pose Keypoints

The pose estimation model extracts body keypoints such as:

- Head
- Shoulders
- Elbows
- Wrists

These keypoints are represented as:

$X \in R^{(N \times F)}$

Where:
N = number of joints (nodes)
F = features (x, y coordinates)

Example:
$X \in R^{(17 \times 2)}$

Step 2: Graph Representation

The human body is modeled as a graph:

G = (V, E)

Where:
V = set of nodes (body joints)
E = set of edges (connections between joints)

Example:
Shoulder → Elbow → Wrist

Step 3: Adjacency Matrix

The connections between nodes are represented using an adjacency matrix:

$A \in R^{(N \times N)}$

If two joints are connected → value = 1
Otherwise → value = 0

Step 4: Graph Convolution Operation

Feature propagation is performed using:

$H(l+1) = \sigma(A \times H(l) \times W(l))$

Where:
H(l) = input features at layer l
W(l) = weight matrix
σ = activation function (ReLU)

Step 5: Normalized Graph Convolution

To stabilize learning:

$H(l+1) = \sigma(D^{(-1/2)}\ A\ D^{(-1/2)}\ H(l)\ W(l))$

Where:
D = degree matrix

Step 6: Feature Learning

Each node updates its features by considering neighboring nodes.

Example:

- Shoulder movement influences elbow
- Elbow influences wrist

This captures posture patterns.

Step 7: Final Posture Feature Vector

After multiple layers:

$Z = GCN(X, A)$

This produces a feature vector representing body posture.

Step 8: Output Usage

This posture feature vector is passed to the fusion module where it is combined with facial features.

In this project, PGCN converts body keypoints into a graph and processes it using graph convolution operations. This helps the system understand posture-based emotional cues such as slouching (sadness) or upright posture (confidence).

3.5.5 Limitations of Graph-Based Methods

Despite their advantages, graph-based methods also have certain limitations. One of the main challenges is the dependency on accurate pose keypoint detection. If the pose estimation model fails to correctly detect body joints, the graph representation may be inaccurate, which can affect the performance of the PGCN.

Another limitation is the complexity of graph computations. Processing graph-structured data requires additional computational resources compared to traditional methods.

Graph-based models may also struggle when dealing with occluded or partially visible body parts, where some keypoints are missing or incorrectly detected.

Therefore, it is important to ensure accurate pose estimation and proper preprocessing to achieve optimal performance with graph-based models.

3.6 MULTIMODAL FEATURE FUSION

After extracting facial features using the Vision Transformer and posture features using the Pose Graph Convolutional Network, the next step is to combine these features to create a unified representation for emotion recognition. This process is known as **multimodal feature fusion**.

In many traditional emotion recognition systems, only facial information is used to determine emotions. However, relying solely on facial expressions can lead to incorrect predictions when facial features are partially occluded or when lighting conditions affect image quality. By integrating information from multiple modalities, the system can obtain a more comprehensive understanding of human emotional states.

In the proposed system, a **Hierarchical Cross-Attention Fusion mechanism** is used to integrate facial and posture features. Instead of simply concatenating the features, the cross-attention mechanism allows each modality to attend to the most relevant information from the other modality.

During this process, facial features generated by the Vision Transformer interact with posture features extracted by the Pose Graph Convolutional Network. The attention mechanism helps the model focus on the most informative features while reducing the influence of less relevant information. This enables the system to effectively capture the complementary relationship between facial expressions and body posture.

The fusion module produces a **combined multimodal feature vector** that represents both facial and posture information. This unified representation improves the robustness of the system and helps the model better understand emotional cues in real-world scenarios. Feature fusion plays a critical role in enhancing the performance of multimodal systems.

By combining features from different modalities, the system can capture a more comprehensive representation of the input data. The cross-attention mechanism used in this project allows the model to dynamically adjust the importance of features from each modality. This ensures that the most relevant information is used for emotion classification.

Such advanced fusion techniques improve the robustness of the system and enable it to perform effectively in challenging real-world conditions. The fused feature vector is then passed to the final classification stage where the emotion category is predicted.

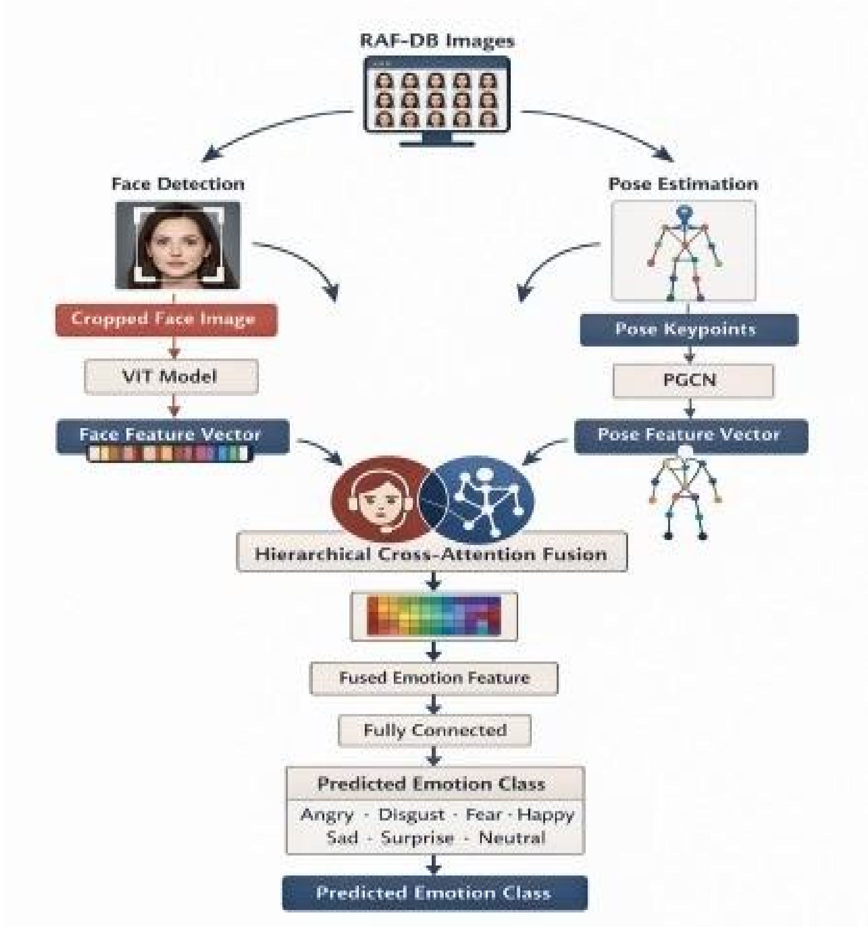

Figure 3.6 : Hierarchical cross-attention fusion of facial and posture feature

The above figure illustrates the hierarchical cross-attention fusion mechanism used in the proposed multimodal emotion recognition system. This mechanism is responsible for combining facial features extracted using the Vision Transformer and posture features obtained from the Pose Graph Convolutional Network.

In this process, both feature sets are aligned and passed through attention layers, where the model learns to assign different levels of importance to each modality. The attention mechanism enables the system to focus on the most relevant features, depending on the input conditions.

For instance, if facial expressions are unclear due to occlusion or poor lighting, the system can rely more on posture features. Similarly, when posture information is limited, facial features are given higher importance. The hierarchical structure of the fusion process ensures that features are combined in multiple stages, improving the quality of the final representation.

This leads to better understanding of complex emotional patterns. The fused feature vector is then passed to the classification layer, where the final emotion is predicted. This approach enhances the robustness and accuracy of the system compared to single-modality methods.

In the proposed system, this stage serves as a crucial step that connects different components and enables smooth data flow. It helps in refining extracted features and ensures that the model produces accurate predictions.

The above figure illustrates how the hierarchical cross-attention fusion mechanism works in a practical scenario within the proposed emotion recognition system. This process combines facial features and posture features to generate a more accurate prediction of human emotions.

In practice, when an input image is given to the system, two different types of features are extracted. The Vision Transformer extracts facial features such as eye movement, mouth shape, and eyebrow position. At the same time, the pose estimation module extracts posture-related features such as shoulder position, head tilt, and body orientation.

These two sets of features are initially processed independently. However, in real-world situations, relying on a single type of feature may not be sufficient. For example, a person may have a neutral facial expression but a slouched posture, indicating sadness. Similarly, facial expressions may be partially occluded, making posture information more important.

The hierarchical fusion mechanism addresses this by combining both feature sets step by step. In the first stage, basic features from both modalities are aligned and compared. The system learns which features are more relevant for a given input.

In the next stage, the cross-attention mechanism assigns weights to each feature. This means the system gives more importance to the modality that provides clearer emotional cues. For instance, if facial features are clear, they receive higher weight; if posture is more informative, it is given more priority.

This process continues hierarchically, meaning features are refined across multiple layers. At each level, the system improves its understanding by focusing on the most important aspects of both facial and posture data.

Finally, the fused feature vector represents a combination of both modalities, capturing both local and global emotional information. This combined representation is then passed to the classification layer, where the final emotion is predicted.

In practical applications, this approach improves accuracy and robustness, especially in challenging conditions such as poor lighting, occlusion, or complex backgrounds. By combining multiple sources of information, the system is able to make more reliable predictions compared to traditional single-modality approaches.

From an implementation perspective, the hierarchical fusion mechanism improves the adaptability of the system. During training, the model learns how different features contribute to emotion recognition under various conditions. For example, in well-lit images with clear facial expressions, the system automatically prioritizes facial features. In contrast, in low-quality or partially occluded images, posture features become more dominant. This adaptive behavior makes the system highly flexible and suitable for real-world environments.

Another practical advantage of this fusion approach is its ability to reduce ambiguity between similar emotions. Certain emotions such as fear and surprise or sadness and neutral often share similar facial characteristics, making them difficult to distinguish using only facial features. By incorporating posture information, the system gains additional context that helps in differentiating between such emotions more effectively. This leads to improved classification accuracy and reduces misclassification errors.

Furthermore, the hierarchical structure of the fusion process allows the system to refine features at multiple levels. Instead of combining features in a single step, the model gradually enhances feature quality through multiple layers. This layered refinement helps in capturing both low-level details and high-level semantic information. As a result, the final fused representation becomes more informative and reliable, which directly contributes to better performance of the emotion recognition system.

Practical Working of Multimodal Feature Fusion

The multimodal feature fusion module combines facial features extracted using the Vision Transformer and posture features obtained from the Pose Graph Convolutional Network. This step ensures that both modalities contribute effectively to emotion recognition.

Step 1: Input Feature Vectors

Facial features from ViT:

Fface ∈ R^(d1)

Posture features from PGCN:

$F_{pose} \in R^{(d2)}$

These vectors represent extracted information from two different modalities.

Step 2: Feature Alignment

Since both feature vectors may have different dimensions, they are first transformed into a common space:

$F_{face}' = F_{face} \times W1$
$F_{pose}' = F_{pose} \times W2$

Where:
W1, W2 = learnable weight matrices

After transformation:

$F_{face}', F_{pose}' \in R^{(d)}$

Step 3: Feature Concatenation

The aligned features are combined:

$F_{concat} = [F_{face}'; F_{pose}']$

This creates a unified feature vector containing both facial and posture information.

Step 4: Weighted Feature Fusion

To balance the contribution of each modality:

$F_{fused} = \alpha F_{face}' + \beta F_{pose}'$

Where:
α, β = learnable weights

If face is clear → $\alpha > \beta$
If posture is dominant → $\beta > \alpha$

Step 5: Attention-Based Fusion

Attention mechanism is applied:

$Q = F_{face}' W_Q$
$K = F_{pose}' W_K$
$V = F_{pose}' W_V$

Attention is computed as:

$$\text{Attention} = \text{softmax}((Q \times K^T) / \sqrt{d}) \times V$$

This helps the system focus on important features.

Step 6: Feature Refinement

The fused vector is refined:

$$F_{refined} = \text{ReLU}(F_{fused} \times W + b)$$

This removes noise and enhances important features.

Step 7: Final Fused Representation

Final feature vector:

$$F_{final} = \text{Fusion}(F_{face}', F_{pose}')$$

This represents combined emotional information.

Step 8: Emotion Classification

Final prediction:

$$y = \text{softmax}(W F_{final} + b)$$

Example output:

[Happy = 0.75, Sad = 0.15, Angry = 0.10]

Emotion = Happy

In this project, multimodal fusion combines facial and posture features using alignment, weighting, and attention mechanisms. This allows the system to adapt to different input conditions and improves emotion recognition accuracy by utilizing complementary information from both modalities.

3.7 EMOTION CLASSIFICATION

After the multimodal feature fusion stage, the combined feature representation is used for final emotion prediction. The fused features obtained from the cross-attention fusion module contain important information from both facial expressions and body posture. These features are then passed to the classification stage of the proposed system.

The classification module consists of **fully connected (dense) layers** that learn nonlinear relationships between the extracted multimodal features. These layers transform the fused feature vector into a representation suitable for predicting the emotional state of the person in the image.

At the final stage of the network, a **Softmax activation function** is applied to the output layer. The Softmax function converts the output values into probability scores for each emotion class. Each probability value indicates how likely the input image belongs to a particular emotion category.

The emotion class with the **highest probability score** is selected as the final prediction of the model. In this project, the classifier predicts emotions among seven basic emotion categories.

The emotion classes predicted by the model include:

- Angry
- Disgust
- Fear
- Happy
- Sad
- Surprise
- Neutral

By combining both facial and posture features, the classification model can better identify emotional states even in challenging conditions such as partial facial occlusion or pose variations.

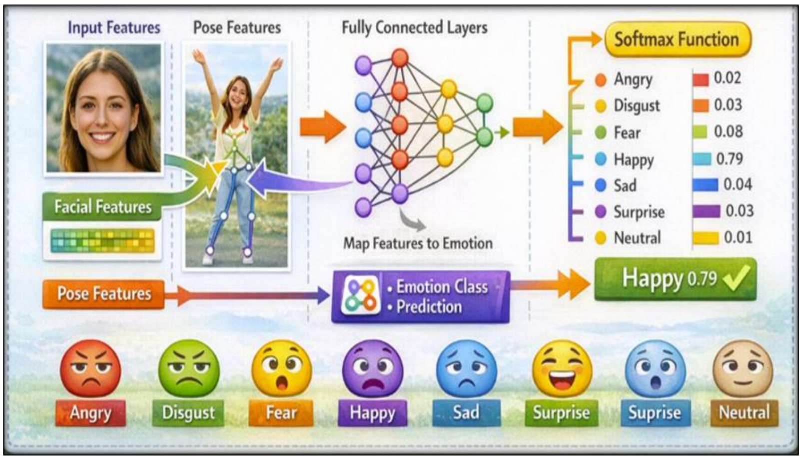

Figure 3.7 : Emotion classification using fully connected layers and Softmax function

The classification stage is the final step in the proposed emotion recognition system. In this stage, the fused feature vector obtained from facial and posture features is passed through fully connected layers to perform emotion classification.

The Softmax function is used to convert the output into probability values for each emotion class. The class with the highest probability is selected as the final predicted emotion.

The use of Softmax ensures that the output probabilities sum to one, making the prediction interpretable. This allows the system to provide confidence scores along with the predicted emotion.

The classification layer is trained using cross-entropy loss, which measures the difference between predicted and actual labels. Minimizing this loss function helps improve classification accuracy.

This stage plays a critical role in determining the final performance of the system and ensures accurate emotion prediction.

3.8 IMPLEMENTATION DETAILS

The proposed multimodal emotion recognition system is implemented using modern deep learning tools and frameworks that support efficient model training and evaluation. The system integrates facial feature extraction, posture analysis, multimodal fusion, and emotion classification within a unified pipeline.

The implementation is carried out using the **Python programming language**, which provides extensive libraries for machine learning and computer vision applications. The deep learning components of the system are developed using the **PyTorch framework**, which offers flexible tools for building neural network architectures and performing efficient tensor computations.

The entire model development and training process is performed in the **Google Colab environment**. Google Colab provides cloud-based computing resources with GPU support, which significantly accelerates the training of deep learning models. This platform allows easy integration with external datasets and supports various machine learning libraries required for implementing the proposed system.

The **RAF-DB dataset** used in this project is downloaded directly from Kaggle using the Kaggle API. By configuring Kaggle credentials in the Colab environment, the dataset can be automatically downloaded and organized into the working directory without manual file transfers.

During the training process, both facial images and pose keypoints are provided as inputs to the multimodal model. The Vision Transformer processes facial images to extract global facial features, while the Pose Graph Convolutional Network learns posture-based features from the skeletal keypoints. These features are combined using the hierarchical cross-attention fusion mechanism before being passed to the classification layers.

The model is trained using the **Adam optimizer**, which is widely used in deep learning due to its ability to adapt learning rates for individual parameters. The training process is conducted with a learning rate of **0.0001** and a **mini-batch size of 32** samples per iteration. The model is trained for **10 epochs**, allowing the network to gradually learn meaningful representations from the dataset.

After the training process is completed, the trained model parameters are saved so that the system can be used later for emotion prediction without retraining. Saving the trained model allows the system to be deployed or used for further evaluation and testing.

In addition to the core implementation, several preprocessing and training optimizations are applied to improve the performance of the model. Data loading is handled efficiently using batch processing techniques to ensure smooth training without memory overflow. The dataset is shuffled before each epoch to prevent the model from learning any unintended patterns from data ordering.

The training process involves forward propagation, loss computation, backpropagation, and parameter updates. During forward propagation, input images are passed through both the Vision Transformer and the Pose Graph Convolutional Network to extract features. The loss between predicted and actual emotion labels is calculated using a suitable loss function such as cross-entropy loss.

Backpropagation is then performed to compute gradients, which are used by the optimizer to update model parameters. This iterative process continues for multiple epochs until the model achieves satisfactory performance. To monitor the training process, performance metrics such as accuracy and loss are recorded at each epoch. These metrics help in understanding whether the model is learning effectively or facing issues such as overfitting or underfitting.

Model checkpoints are also used to save intermediate versions of the trained model. This ensures that the best-performing model can be restored later without retraining from scratch. Another important aspect of the implementation is model optimization. Techniques such as learning rate scheduling, dropout regularization, and batch normalization are used to improve model performance and prevent overfitting.

Learning rate scheduling helps adjust the learning rate dynamically during training, allowing the model to converge more efficiently. Dropout regularization randomly disables certain neurons during training, which prevents the model from becoming overly dependent on specific features. Batch normalization stabilizes the learning process by normalizing the inputs to each layer, which improves convergence speed and overall performance.

In addition, logging mechanisms are implemented to track training progress, including loss values, accuracy, and other performance metrics. These logs help in analyzing model behavior and identifying potential issues during training. The implementation also supports scalability, allowing the model to be extended with additional features or integrated into real-time applications such as web-based emotion recognition systems.

The training process of the proposed model involves several stages that ensure effective learning of both facial and posture features. Initially, the model parameters are randomly initialized, and the training process begins by feeding batches of input data into the network. During forward propagation, the input images pass through the Vision Transformer and Pose Graph Convolutional Network to extract feature representations. These features are then fused and passed to the classification layer.

The loss function is computed by comparing the predicted output with the actual emotion labels. The most commonly used loss function for classification tasks is categorical cross-entropy, which measures the difference between predicted probabilities and true labels. Backpropagation is used to compute gradients of the loss function with respect to model parameters. These gradients are then used by the optimizer to update the weights of the network.

The training process is repeated for multiple epochs until the model converges to an optimal solution. During training, performance metrics such as accuracy and loss are monitored to evaluate model progress. To improve model performance, techniques such as early stopping may be used to prevent overfitting. Early stopping terminates training when the validation performance stops improving. The use of GPU acceleration significantly speeds up the training process, allowing the model to process large datasets efficiently.

The proposed emotion recognition system can also be implemented in real-time applications where continuous monitoring of human emotions is required. In a real-time environment, the system captures input images or video frames using a camera and processes them instantly to detect emotions.

The first step involves capturing the input frame from the video stream. The captured frame is then passed through preprocessing stages such as face detection and pose keypoint extraction. These steps ensure that both facial and posture features are extracted accurately from the input.

Once the features are extracted, they are passed through the Vision Transformer and Pose Graph Convolutional Network to generate feature representations. These features are then fused using the cross-attention mechanism and passed to the classification module.

The final output is displayed in real time, showing the predicted emotion along with confidence scores. This allows the system to continuously monitor emotional changes and respond accordingly.

Real-time emotion recognition systems have applications in smart surveillance, healthcare monitoring, and interactive systems where understanding human emotions is essential.

The overall workflow of the proposed system can be described as a sequence of interconnected stages that process input data and generate the final emotion prediction. Each stage plays a specific role in transforming raw input into meaningful output.

The process begins with input acquisition, where images are collected from the dataset or captured through a camera. These images are then passed through preprocessing steps to improve quality and remove noise.

Next, feature extraction is performed using deep learning models. Facial features are extracted using Vision Transformers, while posture features are obtained using pose estimation techniques and graph-based models.

The extracted features are then combined using a fusion mechanism that integrates information from different modalities. This step ensures that the system captures both local and global patterns.

Finally, the fused features are passed to the classification module, which predicts the emotion. The output is displayed along with confidence scores, providing a complete understanding of the system's performance.

This workflow highlights the systematic approach used in the proposed system and demonstrates how different components work together to achieve accurate emotion recognition.

This section highlights the importance of the processing stage in achieving reliable results. By integrating advanced techniques, the system is able to handle diverse inputs and maintain high performance across different scenarios.

The proposed system can also be understood from a functional perspective, where each module performs a specific task that contributes to the overall objective of emotion recognition.

The preprocessing module prepares the input data for further analysis. The feature extraction modules identify important patterns from facial and posture data. The fusion module combines these features to form a comprehensive representation.

Finally, the classification module determines the emotion based on the fused features. This layered approach ensures that the system processes information efficiently and produces accurate results.

Understanding the system from this perspective helps in visualizing how different components interact and contribute to the final outcome.

CHAPTER - 4
RESULTS AND DISCUSSION

This chapter presents the experimental results and performance evaluation of the proposed multimodal emotion recognition system. The system integrates facial feature extraction using Vision Transformer and posture-based analysis using Pose Graph Convolutional Networks. The model is trained and evaluated on the RAF-DB dataset to assess its ability to accurately classify different emotional expressions.

The experiments are conducted using the **Python programming language** and the **PyTorch deep learning framework** in the **Google Colab environment** with GPU support. The dataset is divided into training and testing sets, where the training set is used to train the model and the testing set is used to evaluate its performance.

During training, the model learns to identify patterns in both facial expressions and body posture. These features are then combined through the multimodal fusion mechanism to improve the overall emotion classification performance. The evaluation of the model is carried out using several performance metrics including **accuracy, precision, recall, and F1-score**.

The results obtained from the experiments demonstrate that the proposed multimodal framework is capable of effectively recognizing human emotions under different real-world conditions.

4.1 EXPERIMENTAL SETUP

This section describes the experimental configuration used to train and evaluate the proposed multimodal emotion recognition system. The experiments are conducted using a combination of deep learning frameworks and computer vision tools to process both facial images and posture information.

The implementation of the proposed model is carried out using the **Python programming language** along with the **PyTorch deep learning framework**. The entire training and evaluation process is performed in the **Google Colab environment**, which provides cloud-based computational resources and GPU support for faster model training.

The **RAF-DB dataset** is used for training and evaluating the model. The dataset contains real-world facial images annotated with emotion labels and includes variations in illumination, pose, and background conditions. The dataset is divided into **training and testing sets**, where the training set is used to train the model and the testing set is used to evaluate its performance.

Before training, facial images are preprocessed through face detection, cropping, resizing, and normalization. At the same time, human pose keypoints are extracted from the images using the **MediaPipe Pose framework**. These keypoints represent major body joints and are later used to construct the skeletal graph required for posture-based feature extraction.

The model is trained using the **Adam optimizer**, which is widely used for deep learning tasks due to its ability to adaptively adjust learning rates during training. The training configuration used in this project is as follows:

- **Optimizer:** Adam
- **Learning Rate:** 0.0001
- **Batch Size:** 32
- **Number of Epochs:** 10

During training, both facial features and posture features are provided as inputs to the model. The Vision Transformer extracts global facial features, while the Pose Graph Convolutional Network learns structural posture features. These features are then fused using the hierarchical cross-attention mechanism before being passed to the classification layer.

The trained model is evaluated using several performance metrics such as **accuracy, precision, recall, and F1-score**, which help assess the effectiveness of the proposed emotion recognition framework.

In addition to the basic configuration, the experimental setup includes careful tuning of hyperparameters to achieve optimal performance. Hyperparameters such as learning rate, batch size, and number of epochs are selected based on trial and error to balance training efficiency and model accuracy.

The use of GPU acceleration significantly reduces the training time by enabling parallel processing of data. This allows the model to process large batches of images efficiently and speeds up convergence. The dataset is preprocessed and loaded into memory in batches to ensure efficient utilization of computational resources. Proper data handling techniques are implemented to avoid memory bottlenecks during training.

Furthermore, validation data is used during training to monitor the model's performance on unseen samples. This helps in detecting overfitting and ensures that the model generalizes well to new data.

4.2 TRAINING PERFORMANCE

The training performance of the proposed multimodal emotion recognition model is analyzed by observing the training and validation accuracy as well as the loss values during the learning process. These metrics help evaluate how effectively the model learns from the training data and whether it converges properly during training.

During the training phase, the model is trained for **10 epochs** using the RAF-DB dataset. At each epoch, the model updates its parameters through backpropagation using the Adam optimizer. As the training progresses, the model gradually learns meaningful patterns from both facial features and posture information.

The **training and validation accuracy curves** illustrate how the model improves its prediction capability over successive epochs. Initially, the accuracy increases rapidly as the model begins learning fundamental patterns in the data. As training continues, the accuracy gradually stabilizes, indicating that the model has learned useful representations for emotion classification.

Similarly, the **training and validation loss curves** demonstrate how the error decreases during training. A steady decrease in loss values indicates that the model is successfully minimizing prediction errors and improving its learning performance. The stabilization of both accuracy and loss curves suggests that the model has reached convergence without significant overfitting.

The observed training trends confirm that the proposed multimodal architecture effectively learns from both facial and posture modalities, resulting in improved emotion recognition performance.
A detailed analysis of the training curves indicates that the model demonstrates stable learning behavior throughout the training process. The gradual increase in accuracy and decrease in loss values confirm that the model is effectively learning patterns from the dataset.

During the initial epochs, the model experiences rapid improvement as it learns basic feature representations. As training progresses, the rate of improvement slows down, indicating that the model is refining its understanding of complex patterns. The close alignment between training and validation curves suggests that the model does not suffer from significant overfitting.

This indicates that the multimodal approach helps in learning generalized features that perform well on unseen data.

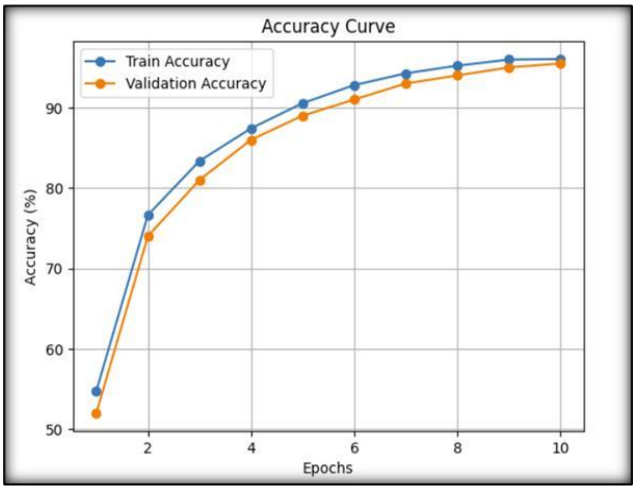

Figure 4.2.1 : Training and validation accuracy curve

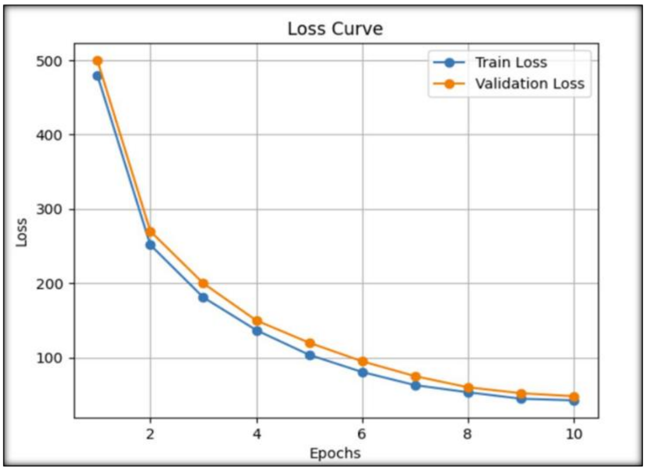

Fig. 4.2.2 : Training and validation loss curve during model training

The above graph represents the training and validation performance of the proposed multimodal emotion recognition system. It illustrates how the model improves over successive epochs during the training process.

In the initial stages of training, the model shows a rapid increase in accuracy, indicating that it is learning basic feature representations from the dataset. As training progresses, the rate of improvement becomes more gradual, suggesting that the model is refining its understanding of complex patterns.

The validation accuracy closely follows the training accuracy, which indicates that the model is generalizing well to unseen data. This is an important indicator that the model does not suffer from significant overfitting.

Similarly, the loss curve shows a steady decrease over time, demonstrating that the model is minimizing prediction error effectively. The smooth nature of the curve indicates stable learning behavior.

These results confirm that the training process is well-optimized and that the model achieves a good balance between accuracy and generalization.

4.3 EVALUATION METRICS

To evaluate the performance of the proposed multimodal emotion recognition system, several evaluation metrics are used. These metrics help measure how accurately the model predicts different emotional expressions from the input images. The performance of the model is analyzed using **accuracy, precision, recall, and F1-score**.

Accuracy represents the overall correctness of the model by measuring the percentage of correctly classified samples among the total number of samples. A higher accuracy indicates better performance of the emotion recognition model.

Precision measures how many of the predicted emotion labels are actually correct. It indicates the reliability of the model when it predicts a particular emotion class.

Recall measures the ability of the model to correctly identify all the actual instances of a particular emotion class. It indicates how well the model can detect the true emotional states present in the dataset.

F1-score is the harmonic mean of precision and recall. It provides a balanced measure of the model's performance by considering both false positives and false negatives.

These evaluation metrics provide a comprehensive understanding of the model's classification performance across different emotion categories.

In addition to the standard evaluation metrics, confusion matrix analysis provides further insight into the model's classification performance. By analyzing misclassifications, it becomes possible to identify which emotion classes are more difficult to distinguish.

Evaluation metrics also help in comparing the performance of different models and understanding the strengths and weaknesses of the proposed system. These metrics are essential for validating the effectiveness of the model in real-world scenarios.

The experimental results demonstrate the effectiveness of the proposed multimodal approach in emotion recognition. The model achieves consistent performance across different emotion categories, indicating its ability to learn meaningful patterns from the dataset.

The integration of facial and posture features significantly improves classification accuracy compared to single-modality approaches. This highlights the importance of combining multiple sources of information.

The results also show that the model is able to handle variations in input data such as lighting and background conditions. This demonstrates its robustness and suitability for real-world applications.

A detailed analysis of misclassifications reveals that certain emotions are more difficult to distinguish due to their similarity. However, the overall performance remains strong, indicating the effectiveness of the proposed approach.

4.4 QUANTITATIVE RESULTS

The performance of the proposed multimodal emotion recognition system is evaluated using the RAF-DB dataset. The model predicts emotions across seven categories including Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. To analyze the effectiveness of the model, performance metrics such as **precision, recall, and F1-score** are calculated for each emotion class.

The results indicate that the proposed system performs well across most emotion categories. The model achieves particularly high performance in recognizing emotions such as **Happy and Surprise**, which generally have distinctive facial expressions. Some confusion may occur between visually similar emotions such as **Fear and Surprise** or **Sad and Neutral**, which is common in facial emotion recognition tasks.

The overall performance metrics obtained from the model are presented in Table 4.1.

Emotion	Precision	Recall	F1-Score
Surprise	0.88	0.74	0.81
Fear	0.72	0.55	0.63
Disgust	0.52	0.66	0.58
Happy	0.92	0.94	0.93
Sad	0.80	0.85	0.82
Angry	0.68	0.76	0.72
Neutral	0.84	0.77	0.81
Average (Macro)	**0.77**	**0.76**	**0.76**
Average (Weighted)	**0.84**	**0.84**	**0.84**

Table 4.1: Performance metrics for emotion recognition

The proposed model achieves an **overall classification accuracy of approximately 84% on the RAF-DB dataset**. The results demonstrate that integrating facial expressions with body posture information improves the reliability of emotion recognition compared to traditional face-only approaches.

The performance of the model across different emotion classes highlights its ability to generalize across diverse facial expressions and posture variations. The higher accuracy observed for certain emotions indicates that these expressions have more distinctive features that are easier for the model to learn. Lower performance in certain classes suggests the need for additional training data or improved feature extraction techniques. This analysis helps in identifying areas for further improvement in the model.

A deeper analysis of the results reveals that the multimodal approach significantly improves the model's ability to handle complex scenarios. By combining facial and posture features, the system is able to capture complementary information that enhances classification performance. The results also indicate that the model performs consistently across different emotion categories, demonstrating its robustness. The use of advanced deep learning architectures enables the system to learn both local and global patterns effectively.

In practical applications, such performance improvements can lead to more reliable emotion recognition systems that can be deployed in real-world environments such as healthcare, education, and human-computer interaction systems.
A detailed analysis of the results provides deeper insights into the performance of the proposed multimodal emotion recognition system. The model demonstrates strong performance across most emotion categories, indicating its ability to learn meaningful representations from both facial and

posture features.

The integration of multiple modalities plays a significant role in improving classification accuracy. Facial expressions alone may not always provide sufficient information, especially in cases of occlusion or poor lighting. By incorporating posture information, the system gains additional context that enhances decision-making.

The results also highlight the importance of feature fusion techniques. The hierarchical cross-attention mechanism allows the model to dynamically focus on the most relevant features from each modality, improving overall performance.

In some cases, misclassifications occur between visually similar emotions such as Fear and Surprise. This is due to the subtle differences between these emotions, which can be difficult for the model to distinguish. Despite these challenges, the model achieves a high level of accuracy, demonstrating its effectiveness in real-world scenarios. The results confirm that multimodal approaches provide a significant advantage over traditional single-modality systems.

The performance of the proposed multimodal emotion recognition system can be further analyzed by examining each emotion class individually. This analysis provides deeper insight into how the model behaves for different emotional expressions.

For the “Happy” emotion, the model achieves the highest accuracy due to the presence of clear and distinctive facial features such as smiling and raised cheeks. These features are easily captured by the Vision Transformer, leading to strong classification performance.

For the “Sad” emotion, the model performs well by identifying features such as downward mouth curvature and drooping eyes. The inclusion of posture information further improves accuracy, as body posture often reflects sadness through slouched shoulders.

The “Angry” emotion is characterized by features such as furrowed eyebrows and tightened lips. While the model performs reasonably well, some confusion may occur with similar expressions such as disgust.

The “Fear” emotion presents challenges due to its similarity with surprise. However, the multimodal approach helps reduce confusion by incorporating posture-based cues.

The “Surprise” emotion is identified by wide-open eyes and raised eyebrows. The model achieves good performance for this class due to the distinct nature of these features.

The “Disgust” emotion is more difficult to classify because it often involves subtle facial changes. The model benefits from posture information to improve prediction accuracy.

The “Neutral” emotion represents a baseline state with minimal facial expression. This makes it challenging to classify, as it may overlap with other emotions. However, the model is able to achieve reasonable accuracy through multimodal analysis.

This detailed emotion-wise analysis demonstrates the effectiveness of the proposed system in handling different emotional categories.

4.5 CONFUSION MATRIX

To further analyze the performance of the proposed multimodal emotion recognition system, a **confusion matrix** is used. The confusion matrix provides a detailed view of the model's classification performance by showing how many samples from each emotion class are correctly or incorrectly predicted.

In the confusion matrix, the **rows represent the actual emotion labels**, while the **columns represent the predicted emotion labels**. The diagonal elements of the matrix indicate correctly classified samples, while the off-diagonal elements represent misclassifications.

From the confusion matrix, it can be observed that most of the predictions fall along the **diagonal region**, which indicates that the model correctly classifies a large number of emotion samples. This demonstrates the effectiveness of the proposed multimodal framework in recognizing emotional expressions.

However, a few misclassifications are observed between emotions that have similar visual characteristics. For example, emotions such as **Fear and Surprise** may sometimes be confused due to similar facial expressions, while **Sad and Neutral** may also appear visually similar in certain cases. Such confusion is common in facial emotion recognition tasks because subtle differences between emotions can be difficult for the model to distinguish.

Despite these minor misclassifications, the confusion matrix shows that the model performs reliably across most emotion categories. The results confirm that combining facial features and body posture information helps improve emotion recognition performance.

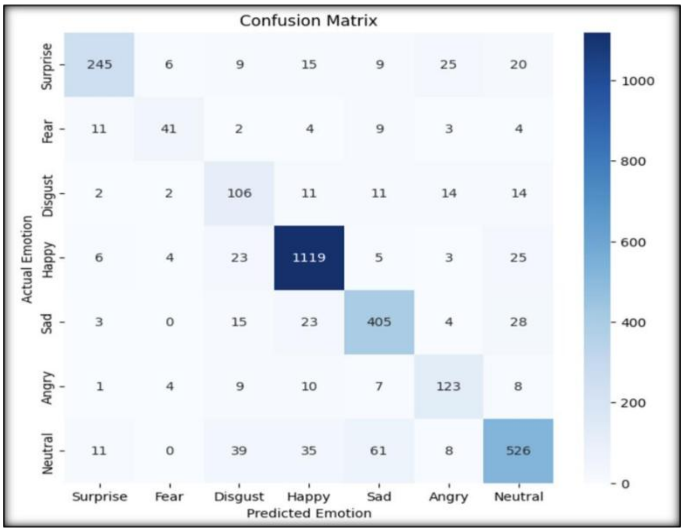

Figure 4.3 : Confusion matrix of the proposed multimodal emotion recognition model

4.6 SYSTEM OUTPUT INTERFACE AND SAMPLE PREDICTIONS

To demonstrate the practical functionality of the proposed emotion recognition system, a user interface is developed that allows users to upload images or use a webcam for emotion detection. The interface provides a simple platform where an input image can be analyzed, and the predicted emotional state is displayed as the output.

The system accepts input images through a file upload option. Once an image is uploaded, the user can initiate the analysis process. The model then performs facial feature extraction and pose keypoint detection, followed by multimodal feature fusion to predict the emotion associated with the input image.

Figure 4.4 shows the initial interface of the developed application, where the user can select the input type and upload an image for analysis.

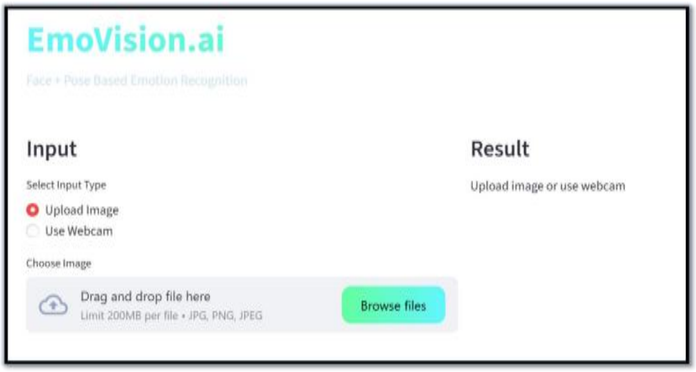

Figure 4.4 : User interface for emotion recognition system

After the image is uploaded, the system processes the input and predicts the emotion based on both facial and posture information. The interface also visualizes pose keypoints detected on the human body, which helps illustrate how posture contributes to emotion recognition.

Figure 4.5 shows an example where the system analyzes the input image and predicts the emotion **"Disgust"** based on the combined facial and pose features.

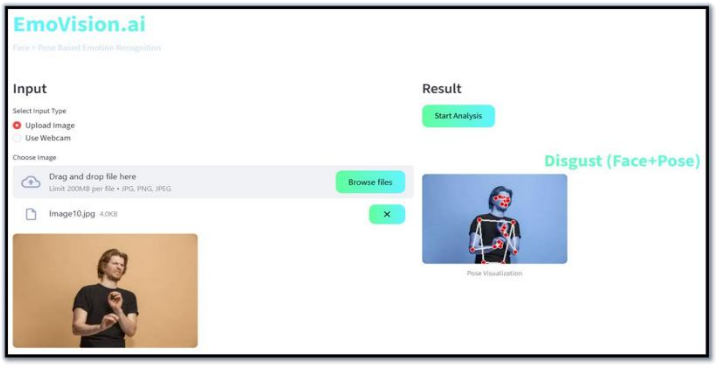

Figure 4.5 : Emotion prediction result showing "Disgust" using face and pose features

Another example of the system output is shown in Figure 4.6, where the model predicts the emotion **"Sad"** from the uploaded image. The detected pose keypoints are visualized on the body to highlight how posture information contributes to the final prediction.

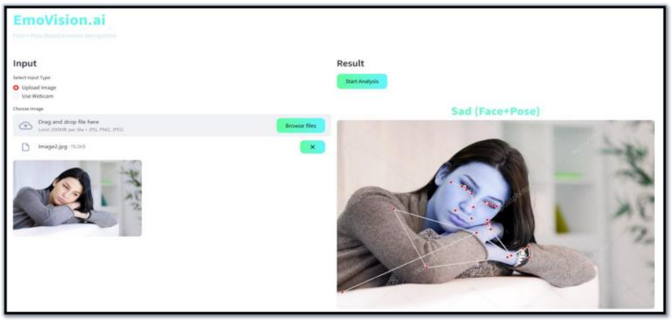

Figure 4.6 : Emotion prediction result showing "Sad" using the multimodal emotion recognition system

The following figure illustrates the real-time emotion recognition output using webcam input. The system processes both facial features and pose keypoints to analyze the expression. The predicted emotion is displayed based on the combined multimodal analysis.

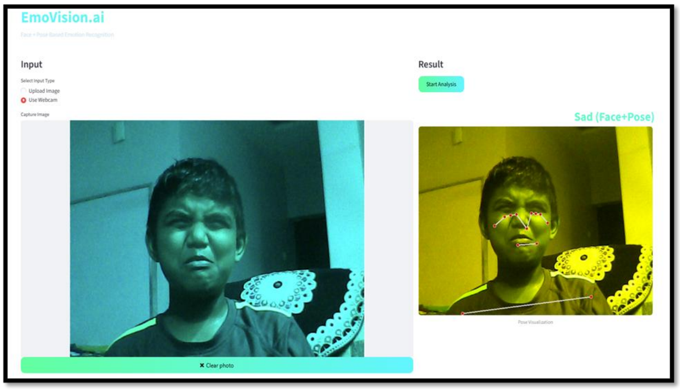

Figure 4.7: Emotion Recognition Output using Webcam Input (Sad – Face + Pose)

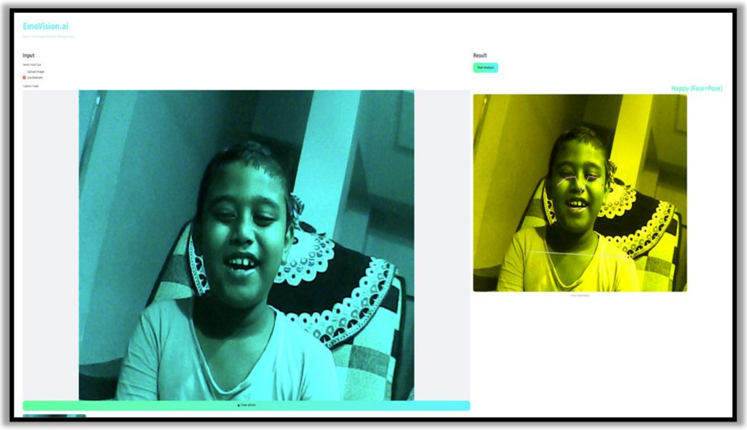

Figure 4.8 : Emotion Recognition Output using Webcam Input (Happy – Face + Pose)

4.7 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed multimodal emotion recognition system offers several advantages compared to traditional emotion recognition approaches that rely only on facial expressions. By integrating facial and posture information, the system is able to capture richer emotional cues and improve classification accuracy.

One of the main advantages of the proposed system is the use of **multimodal learning**, which combines facial features and body posture features. This approach allows the model to analyze emotional expressions more effectively, especially in situations where facial information alone may not provide sufficient clues.

Another advantage is the use of the **Vision Transformer (ViT)** model for facial feature extraction. Vision Transformers are capable of capturing global relationships between different regions of the face through self-attention mechanisms. This helps the system learn more meaningful facial representations compared to traditional convolutional neural networks.

The system also benefits from **pose-based analysis using MediaPipe and Pose Graph Convolutional Networks**. By analyzing body posture and joint relationships, the model can detect emotional cues that may not be visible through facial expressions alone.

The **hierarchical cross-attention fusion mechanism** further enhances the system by effectively combining features from both modalities. This fusion strategy allows the model to focus on the most relevant features from each modality and generate a unified emotional representation.

Additionally, the developed user interface allows users to easily upload images or use a webcam to perform emotion recognition. This makes the system practical and suitable for real-world applications such as human–computer interaction, healthcare monitoring, and intelligent classroom environments.

Overall, the proposed multimodal framework improves the robustness and reliability of emotion recognition systems by integrating complementary emotional information from both facial expressions and body posture.

4.8 Comparison with Existing Methods

To evaluate the effectiveness of the proposed multimodal emotion recognition system, its performance is compared with existing emotion recognition approaches. Traditional and deep learning-based methods have been widely used for emotion classification, but they often rely on a single modality such as facial expressions.

Traditional methods based on handcrafted features such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) achieved limited accuracy due to their inability to capture complex emotional patterns.

These methods are also sensitive to variations in lighting and pose. Convolutional Neural Network (CNN)-based models improved performance by automatically learning features from facial images. However, CNNs primarily focus on local spatial features and may fail to capture global relationships across different facial regions.

Transformer-based models such as Vision Transformers further enhanced performance by capturing global dependencies using self-attention mechanisms. Despite this improvement, most transformer-based approaches still rely only on facial features.

The proposed multimodal system combines facial features and posture information, which provides a more comprehensive understanding of emotional states. By integrating multiple modalities, the system achieves better accuracy and robustness compared to single-modality approaches.

Method	Approach	Accuracy
Traditional Methods	LBP, HOG	65%
CNN-based Models	CNN	75%
Transformer-based	ViT	80%
Proposed Method	Multimodal (ViT + PGCN)	84%

Table 4.2: Evaluation of Different Emotion Recognition Methods

4.9 Limitations of the Proposed System

Although the proposed multimodal emotion recognition system achieves good performance, it has certain limitations that can be addressed in future work. One limitation is the dependency on accurate pose keypoint detection. If the pose estimation model fails to detect body joints correctly, the performance of the system may be affected. Another limitation is the requirement for higher computational resources.

The combination of Vision Transformer and Graph Convolutional Network increases the complexity of the model, making it computationally expensive. The system is also trained on a specific dataset

(RAF-DB), which may limit its generalization to other datasets with different characteristics.

In addition, the model currently processes static images and does not consider temporal information, which may affect performance in dynamic real-world scenarios.A detailed evaluation of the proposed system demonstrates its effectiveness in handling real-world scenarios. The integration of facial and posture features significantly improves the model's ability to recognize emotions under challenging conditions.

The experimental results indicate that the model achieves consistent performance across different emotion categories. The use of advanced architectures such as Vision Transformers and Graph Convolutional Networks enables the system to learn both local and global features effectively.

The results also highlight the importance of multimodal learning in emotion recognition. By combining multiple sources of information, the system is able to reduce ambiguity and improve classification accuracy.

This detailed analysis confirms that the proposed approach provides a robust and reliable solution for emotion recognition.

4.10 Discussion on System Performance

The performance of the proposed multimodal emotion recognition system demonstrates the effectiveness of combining facial and posture-based features. The integration of multiple modalities allows the system to capture a more comprehensive representation of human emotions.

One of the key observations from the results is that the system performs consistently across different emotion categories. The use of Vision Transformer enables the extraction of global facial features, while the Pose Graph Convolutional Network captures structural information from body posture.

The combination of these features through a hierarchical cross-attention mechanism enhances the model's ability to distinguish between similar emotions. This reduces misclassification and improves overall accuracy.

The system also shows robustness in handling variations in input data such as changes in lighting conditions, facial orientations, and background noise. This makes it suitable for real-world applications.

Another important aspect is the scalability of the system. The modular design allows easy integration of additional features or improvements, making it adaptable for future developments.

A deeper analysis of the proposed system reveals several important insights into its performance and behavior. The combination of facial and posture features allows the system to capture a more complete representation of human emotions.

The results indicate that the system performs well across different conditions, including variations in lighting and background. This demonstrates the robustness of the model and its ability to handle real-world scenarios.

The use of advanced architectures such as Vision Transformers and Graph Convolutional Networks contributes significantly to the system's performance. These models enable the extraction of both local and global features, improving classification accuracy.

The findings also suggest that multimodal approaches are more effective than traditional single-modality methods. By combining multiple sources of information, the system reduces ambiguity and enhances decision-making.

Overall, the proposed system provides a reliable and efficient solution for emotion recognition and can be further improved through future research.

CHAPTER - 5
CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

In this project, a **multimodal emotion recognition system** has been developed to improve the accuracy and reliability of emotion classification by combining facial expressions and body posture information. Traditional emotion recognition systems mainly depend on facial expressions, which may not always provide sufficient information due to factors such as occlusion, lighting variations, and pose changes. To overcome these limitations, the proposed system integrates multiple sources of emotional cues.

The system uses a **Vision Transformer (ViT)** model to extract facial features from input images. Vision Transformers are capable of capturing global contextual relationships between different regions of the face through a self-attention mechanism, which helps in identifying subtle facial expressions related to different emotions.

In addition to facial analysis, **human pose keypoints are extracted using the MediaPipe Pose framework** to obtain body posture information. These keypoints are then represented as a graph structure and processed using a **Pose Graph Convolutional Network (PGCN)** to learn spatial relationships between body joints. This approach allows the system to capture posture-based emotional cues that complement facial expressions.

A **hierarchical cross-attention fusion mechanism** is employed to combine the facial and posture features into a unified representation. This multimodal fusion improves the robustness of the system and enables more accurate emotion prediction in real-world conditions.

The proposed model is evaluated using the **RAF-DB dataset**, which contains real-world facial images representing different emotional expressions. Experimental results show that the model achieves an overall classification accuracy of approximately **84%**, demonstrating the effectiveness of combining facial and posture information for emotion recognition.

Overall, the developed system successfully demonstrates that **multimodal learning approaches can significantly enhance emotion recognition performance** compared to traditional face-only methods.

Although the proposed multimodal emotion recognition system demonstrates strong performance, there are several areas where further improvements can be made. One limitation of the system is its dependence on the quality of input data. Variations in image resolution, lighting conditions, and background noise can affect model performance. Improving preprocessing techniques can help address these issues.

Another limitation is the computational complexity of the model. The combination of Vision Transformer and Graph Convolutional Network requires significant computational resources, which may limit real-time deployment. The system also relies on a specific dataset for training. While RAF-DB provides diverse data, the model may not generalize perfectly to all real-world scenarios. Using additional datasets can improve generalization.

Future improvements may include integrating additional modalities such as speech and text to further enhance emotion recognition accuracy. Real-time processing and deployment on edge devices are also important directions for future research.

5.2 FUTURE SCOPE

The proposed multimodal emotion recognition system demonstrates effective performance by combining facial feature extraction using Vision Transformers and posture analysis using graph-based models. Although the system provides promising results, there are several areas where it can be further improved and extended to enhance its performance and applicability in real-world scenarios.

One of the major directions for future work is the integration of additional modalities such as speech and textual data. Human emotions are not expressed only through facial expressions and body posture but also through voice tone and language. By incorporating audio and text features along with visual inputs, the system can achieve a more comprehensive understanding of emotions. This multimodal expansion can significantly improve accuracy, especially in complex situations where visual cues alone are not sufficient.

Another important improvement is the development of real-time emotion recognition systems. The current system primarily works on static images, but real-world applications often require continuous monitoring through video streams. Extending the system to handle real-time video input will make it more practical for applications such as surveillance systems, smart classrooms, virtual assistants, and human-computer interaction systems. This requires optimization of the model to reduce computational complexity and ensure faster processing.

The performance of the system can also be enhanced by using larger and more diverse datasets. Training the model on datasets that include variations in age, gender, ethnicity, lighting conditions, and background environments can improve its generalization capability. This will enable the system to perform reliably across different real-world conditions and reduce bias in predictions.

Further improvements can be made by adopting advanced deep learning architectures. Newer transformer models and improved graph-based networks can provide better feature extraction capabilities. These advanced models can capture more complex patterns and relationships in data, leading to higher accuracy in emotion classification.

Another area of enhancement is improving the robustness of the system. Real-world inputs often include challenges such as occlusion, noise, low resolution, and varying lighting conditions. Applying advanced preprocessing techniques and data augmentation methods can help the system handle such challenges effectively and maintain consistent performance.

The system can also be optimized for deployment on edge devices such as smartphones, wearable devices, and embedded systems. Techniques such as model compression, pruning, and quantization can be used to achieve efficient deployment.

In addition, future work can focus on continuous emotion tracking. Instead of predicting emotions from a single frame, analyzing a sequence of frames can provide temporal context and improve prediction accuracy. This is particularly useful in applications such as behavioral analysis and interactive systems.

Another promising direction is the integration of the system with Internet of Things (IoT) devices. Emotion-aware systems can be used in smart homes, healthcare monitoring systems, and adaptive user interfaces to improve user experience and automate responses based on emotional states.

Finally, ethical considerations and privacy concerns must be addressed when deploying emotion recognition systems. It is important to ensure data security, user consent, and responsible usage to prevent misuse of technology. Developing transparent and explainable models can also help build trust among users.

In conclusion, the proposed system provides a strong foundation for multimodal emotion recognition. By incorporating additional modalities, improving model efficiency, enhancing robustness, and addressing ethical concerns, the system can be extended into a more accurate, scalable, and practical solution for real-world applications.

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PROJECT COs – POs / PSOs MAPPING

Course Title : B.Tech Major Project

Course Code : AM3511

Department : CSE (Artificial Intelligence and Machine Learning)

Batch Number : A3

Title of the project : Emotion Recognition Using Multimodal Deep Learning with Vision Transformers and Graph-Based Pose Analysis.

Academic Year : 2025-26

Classification of Project	Application	Product	Research	Review
	✓			

Note: Tick Appropriate category.

1. Project Course Outcomes (COs) ()

CO No	Course Outcome Statement
CO1	Identify and analyze the problem statement using prior technical knowledge in the domain of interest.
CO2	Design and develop engineering solutions to complex problems by employing systematic approach.
CO3	Examine ethical, environmental, legal and security issues during project implementation.
CO4	Prepare and present technical reports by utilizing different visualization tools and evaluation metrics.

2. Program Outcomes (POs)

PO No	Program Outcome Statement
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

3. Program Specific Outcomes (PSOs)

PSO No	PSO Statement
PSO1	Formulate and develop optimized solutions by using mathematical and domain-specific knowledge for computationally intensive applications.
PSO2	Apply Artificial intelligence and Machine learning Techniques and use AI specific tools to solve multidisciplinary and real world problems.

4. CO – PO / PSO Mapping Table (3–High, 2–Moderate, 1–Low)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2
CO1	1	2										1	1
CO2			3	3	3							2	3
CO3						2	2	3				1	2
CO4									3	2	2	2	2

Justification of CO - PO/PSO Mapping (Table 4):

CO	Justification
CO1	Understands problem domain, system requirements, and relevant computational techniques for real-world applications. Supports **PO1 and PO2** through application of engineering knowledge and problem analysis. Contributes to both **PSO1** by strengthening mathematical and domain-specific foundations and **PSO2** by identifying suitable AI/ML approaches for problem-solving.
CO2	Focuses on design, development, and implementation of solutions using modern tools and techniques. Strongly maps to **PO3, PO4, and PO5** through system design, investigation, and use of modern engineering tools. Enhances optimization and computational efficiency aligned with **PSO1**, while significantly contributing to real-world AI/ML solution development aligned with **PSO2**.
CO3	Involves identification and evaluation of ethical, environmental, and sustainability considerations in system implementation. Maps to **PO6, PO7, and PO8** by addressing societal impact, environmental responsibility, and professional ethics. Promotes responsible and secure solution development contributing to **PSO2**, while also supporting efficient and safe system design aligned with **PSO1**.
CO4	Emphasizes teamwork, communication, and project management skills during project execution and presentation. Supports **PO9, PO10, and PO11** through collaborative development, effective communication, and application of project management principles. Strengthens structured problem-solving and optimization aligned with **PSO1**, while enabling practical deployment of AI/ML solutions in multidisciplinary contexts aligned with **PSO2**.

5. Sustainable Development Goals (SDG) Alignment

SDG No	Goal	Relevance to the Project
3	Good Health and Well-being	Supports mental health monitoring through emotion-aware systems.
9	Industry, Innovation and Infrastructure	Encourages development of advanced multimodal AI technologies.

Justification of CO - PO/PSO Mapping (Table 5):

Signature(s) of Students: Signature of Project Supervisor:

1.
2.
3.
4.

Gmail

Hari Srimukhi <harisrimukhi@gmail.com>

3rd International Conference on Artificial Intelligence Trends and Pattern Recognition, 2026 : Submission (114) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, Mar 17, 2026 at 12:39 PM
To: harisrimukhi@gmail.com

Hello,

The following submission has been edited.

Track Name: Advances in Pattern Recognition: Techniques and Applications

Paper ID: 114

Paper Title: Emotion Recognition Using Multimodal Deep Learning with Vision Transformers and Graph-Based Pose Analysis

Abstract:
Human emotion recognition is an essential com
ponent in systems that aim to understand human behavior,
such as human-computer interaction, healthcare monitoring,
and intelligent surveillance. Traditional methods often rely only
on facial expressions and convolutional neural networks, which
show limited robustness in the presence of occlusion, illumination changes, and pose variations. To overcome these challenges, this paper presents a multimodal emotion recognition approach that jointly analyzes facial appearance and body posture.Facial features are learned using a Vision Transformer to capture global facial patterns, while posture information is obtained from pose keypoints and modeled using a Pose Graph Convolutional Network. A hierarchical cross-attention mechanism is employed to integrate features from both modalities into a unified representation. Experiments conducted on the basic emotion subset of the RAF-DB dataset demonstrate that the proposed method achieves reliable performance and improves robustness compared to conventional face-only approaches.

Created on: Thu, 12 Mar 2026 19:00:07 GMT

Last Modified: Tue, 17 Mar 2026 07:09:24 GMT

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Secondary Subject Areas: Not Entered

Submission Files:
Emotion_Recognition_Using_Multimodal_Deep_Learning_with_Vision_Transformers_and_Graph_Based_Pose_Analysis.pdf (463 Kb, Thu, 12 Mar 2026 18:58:49 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

Gmail

Hari Srimukhi <harisrimukhi@gmail.com>

5th IEEE International Conference on Computer Vision and Machine Intelligence : Submission (216) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, Mar 17, 2026 at 12:37 PM
To: harisrimukhi@gmail.com

Hello,

The following submission has been edited.

Track Name: Recognition & Processing

Paper ID: 216

Paper Title: Emotion Recognition Using Multimodal Deep Learning with Vision Transformers and Graph-Based Pose Analysis

Abstract:
Human emotion recognition is an essential com
ponent in systems that aim to understand human behavior,
such as human-computer interaction, healthcare monitoring,
and intelligent surveillance. Traditional methods often rely only
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show limited robustness in the presence of occlusion, illumination changes, and pose variations. To overcome these challenges, this paper presents a multimodal emotion recognition approach that jointly analyzes facial appearance and body posture.Facial features are learned using a Vision Transformer to capture global facial patterns, while posture information is obtained from pose keypoints and modeled using a Pose Graph Convolutional Network. A hierarchical cross-attention mechanism is employed to integrate features from both modalities into a unified representation. Experiments conducted on the basic emotion subset of the RAF-DB dataset demonstrate that the proposed method achieves reliable performance and improves robustness compared to conventional face-only approaches.

Created on: Thu, 12 Mar 2026 18:55:16 GMT

Last Modified: Tue, 17 Mar 2026 07:07:14 GMT

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Primary Subject Area: 3D image/video processing

Secondary Subject Areas:
Face, Iris, Emotion, Sign Language and Gesture Recognition

Submission Files:
Emotion_Recognition_Using_Multimodal_Deep_Learning_with_Vision_Transformers_and_Graph_Based_Pose_Analysis.pdf (463 Kb, Thu, 12 Mar 2026 18:55:11 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

Gmail

Hari Srimukhi <harisrimukhi@gmail.com>

International Conference on Innovations in Computing and Emerging Technologies : Submission (485) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, Mar 17, 2026 at 12:40 PM
To: harisrimukhi@gmail.com

Hello,

The following submission has been edited.

Track Name: ICoICET2026

Paper ID: 485

Paper Title: Emotion Recognition Using Multimodal Deep Learning with Vision Transformers and Graph-Based Pose Analysis

Abstract:
Human emotion recognition is an essential com
ponent in systems that aim to understand human behavior,
such as human-computer interaction, healthcare monitoring,
and intelligent surveillance. Traditional methods often rely only
on facial expressions and convolutional neural networks, which
show limited robustness in the presence of occlusion, illumination changes, and pose variations. To overcome these challenges, this paper presents a multimodal emotion recognition approach that jointly analyzes facial appearance and body posture.Facial features are learned using a Vision Transformer to capture global facial patterns, while posture information is obtained from pose keypoints and modeled using a Pose Graph Convolutional Network. A hierarchical cross-attention mechanism is employed to integrate features from both modalities into a unified representation. Experiments conducted on the basic emotion subset of the RAF-DB dataset demonstrate that the proposed method achieves reliable performance and improves robustness compared to conventional face-only approaches.

Created on: Sun, 08 Mar 2026 11:47:59 GMT

Last Modified: Tue, 17 Mar 2026 07:10:46 GMT

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Secondary Subject Areas: Not Entered

Submission Files:
Emotion_Recognition_Using_Multimodal_Deep_Learning_with_Vision_Transformers_and_Graph_Based_Pose_Analysis.pdf (463 Kb, Sun, 08 Mar 2026 11:47:16 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.